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MEDISAFE SI PCF USER MANUAL



Quick Links

Cycle Errors
Routine Maintenance
Troubleshooting
Error Messages

Table of Contents

Table of Contents

Medisafe Contact Details

Independent Monitoring

Memorandum of Registration

SI PCF Warranty Registration

Safety Precautions

Training Personnel

Compliance

Product Identity

Manufacturer Information

Intended Use

Performance

Cycle Record

Cycle Failure

Concept Summary

General Safety

Chemicals

Decontamination

Electrical Safety

Heat

Service Contracts & Warranty

Validation Option

Transport and Storage

Marking

Environmental

Lifting

Unpacking

Returns

Enclosures Check List

Robotic PCF S
Standard PCF S
Standard Accessory Kit (MP21424)
Additional Accessories
Approved Chemicals
Baskets
Printer
Installation Requirements
Connecting the SI PCF
Installation Drawings
Front View
Rear View Showing Services
Right Side View Lid Closed
Right Side View Lid Open
Rear View Alternative Drain Connection
Right Side View Alternative Drain Connection
CONNECTING TO THE RS232 PORT
Thermal Printer
Data Logging PC
Menu
Power Up
Authorisation code
Setting Time & Date
Set Time
Set Date
Selecting a program
Standard Program Parameters
Robotic Program Parameters
Chemical Handling
Chemical Consumption
Chemical Priming
Chemical Replacement
Disposal of chemical containers
Baskets

Basket Insertion
Basket Removal
Instruments
Instrument Loading Precautions
Starting a Standard Instrument Cycle
Instrument Loading
Manual Pre Wash
Starting a Robotic Instrument Cycle
Automated Wash Cycle
Cycle Process
Pre-wash
Wash
Final Rinse/Disinfection
Rinse
Cooling
Cycle Complete
Robotic Instrument Preparation
Unload
Re-Wash Cycle
Aborting a Cycle
Cycle Errors
Download Logs
Downloading the Cycle Log
Routine Maintenance
Before Every Cycle
Daily Checks
Weekly checks
Daily Cleaning/ Procedure
Decontamination Guide
Full Decontamination
Pre Transport Procedure
Decontamination Certificate
Error Messages
Other Faults
General Specifications

Applied Standards
EMC
EMC & Electromagnetic Compatibility Data
Electromagnetic Emissions
Electromagnetic Immunity
Separation Distances Between SI PCF and Mobile RF Equipment
Warranty
Service Record
Operator PIN Number Codes
Symbols Glossary
Other ManualsLib Projects

Medisafe®

SI PCF

User Manual

S Series



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Table of Contents

1. INTRODUCTION 7

1-1 Medisafe Contact Details 8
1-2 Independent Monitoring 9
1-3 Memorandum of Registration 10
1-4 SI PCF Warranty Registration 11
1-5 Safety Precautions 13
1-6 Training Personnel 16
1-7 Compliance 17
1-8 Product Identity 17
1-9 Manufacturer Information 17
1-10 Intended Use 18
1-11 Performance 18
1-12 Cycle Record 18
1-13 Cycle Failure 18
1-14 Concept Summary 18
1-15 General Safety 19
1-15-1 Chemicals 19
1-15-2 Decontamination 19
1-15-3 Heat 19
1-16 Electrical Safety 19
1-17 Service Contracts & Warranty 19
1-18 Validation Option 19

2. INSTALLATION 21

2-1 Transport and Storage 22
2-1-1 Marking 22
2-1-2 Environmental 22
2-2 Lifting 22
2-3 Unpacking 22
2-4 Returns 22
2-5 Enclosures Check List 23
2-5-1 Standard PCF S 23
2-5-2 Robotic PCF S 23
2-5-3 Standard Accessory Kit (MP21424) 24
2-6 Additional Accessories 25
2-6-1 Baskets 25
2-6-2 Printer 25
2-6-3 Approved Chemicals 25
2-7 Installation Requirements 26
2-8 Connecting the SI PCF 28
2-9 Installation Drawings 29
2-9-1 Front View 29
2-9-2 Rear View Showing Services 30
2-9-3 Right Side View Lid Closed 31
2-9-4 Right Side View Lid Open 32
2-9-5 Rear View Alternative Drain Connection 33
2-9-6 Right Side View Alternative Drain Connection 34
CONNECTING TO THE RS232 PORT 35

Downloaded from www.Manualslib.com manuals search engine

2-10 Thermal Printer *35*
2-11 Data Logging PC *35*

3. OPERATION **37**

3-1 Menu *38*
3-2 Power Up *39*
3-3 Authorisation code *39*
3-4 Setting Time & Date *39*
3-4-1 Set Time 39
3-4-2 Set Date 39
3-5 Selecting a program *40*
3-5-1 Standard Program Parameters 40
3-5-2 Robotic Program Parameters 41
3-6 Chemical Handling *42*
3-6-1 Chemical Consumption 43
3-6-2 Chemical Replacement 43
3-6-3 Disposal of chemical containers 43
3-6-4 Chemical Priming 43
3-7 Baskets *44*
3-7-1 Basket Removal 44
3-7-2 Basket Insertion 44
3-8 Instruments *44*
3-8-1 Instrument Loading Precautions 44
3-9 Starting a Standard Instrument Cycle *45*
3-10 Starting a Robotic Instrument Cycle *46*
3-10-1 Manual Pre Wash 46
3-10-2 Instrument Loading 46
3-11 Automated Wash Cycle *48*
3-11-1 Cycle Process 48
3-11-2 Pre-wash 49
3-11-3 Wash 49
3-11-4 Rinse 50
3-11-5 Final Rinse/Disinfection 50
3-11-6 Cooling 51
3-11-7 Cycle Complete 51
3-11-8 Unload 51
3-11-9 Robotic Instrument Preparation 51
3-11-10 Re-Wash Cycle 52
3-12 Cycle Errors *53*
3-13 Aborting a Cycle *53*
3-14 Downloading the Cycle Log *54*
3-14-1 Download Logs 54
3-15 Routine Maintenance *55*
3-15-1 Before Every Cycle 56
3-15-2 Daily Checks 56
3-15-3 Weekly checks 56
3-16 Decontamination Guide *57*
3-16-1 Daily Cleaning/ Procedure 57
3-16-2 Full Decontamination 57
3-16-3 Pre Transport Procedure 57
3-16-4 Decontamination Certificate 58

4. TROUBLESHOOTING **59**

Downloaded from www.Manualslib.com manuals search engine

4-1 Error Messages *60*
4-2 Other Faults *62*

5. TECHNICAL **65**

5-1 General Specifications *66*
5-2 Applied Standards *70*

6. APPENDICES **71**

EMC 72
6-1 EMC & Electromagnetic Compatibility Data *72*
6-3 Electromagnetic Emissions *73*
6-4 Electromagnetic Immunity *74*
6-5 Separation Distances Between SI PCF and Mobile RF Equipment *76*
6-6 Warranty *77*
6-7 Service Record *78*
6-8 Operator PIN Number Codes *79*
6-9 Symbols Glossary *81*

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1. Introduction

User Manual



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1-1 Medisafe Contact Details

United Kingdom

(Manufacturing, Sales, Customer Service & Technical Support)

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Unit 1 Twyford Industrial Estate
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www.medisafeinternational.com

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Introduction



Important

Read this entire manual before attempting to install, operate or perform any maintenance on the Sonic Irrigator Parametric Controlled Flow (SI PCF) Ultrasonic Cleaning System. Ensure all appropriate personnel understand the contents of this manual.

This Operating Manual contains important information on proper use and maintenance of the SI PCF Ultrasonic Cleaning System. All personnel involved in the use and maintenance of this equipment must carefully review and comply with the warnings, cautions and instructions contained in this manual. These instructions are important to protect the health and safety of personnel operating the Ultrasonic Cleaning System and should be retained in a conveniently accessible area for quick reference.

Advisory

A listing of the *Safety Precautions* to be observed when operating the SI PCF Ultrasonic Cleaning System is found in Section 1-5 of this manual. Do not operate the equipment until you have become familiar with this information.

1-2 Independent Monitoring

Important Note for the Product Owner

It is advised that an independent monitoring system is fitted in accordance with ISO 15883-1:2009 - Washer-disinfectors.

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1-3 Memorandum of Registration

Important

The person undertaking the installation must complete this Memorandum of Registration. In the event of a Warranty Claim, the information contained within this Registration is your evidence that the product has been correctly installed.

Important Note for the Installer

This *Memorandum of Registration* MUST be completed by the person undertaking installation and witnessed by the owner of the product. For installation instructions and contact information for qualified personnel who can perform the installation, please contact Medisafe (see section 1-1).

Failure to provide this evidence in the event of a warranty claim may result in the claim being rejected (*this does not affect your statutory rights as a customer*).

Owner Name:	
Installation Address:	
Installation Date:	
Model No.:	
Serial No.:	
Software Version:	
Installed by:	
Witnessed by:	

Service and Maintenance Requirements

The SI PCF should be serviced on a regular basis to ensure that the product continues to perform in accordance with its design parameter, see section 6-7 for more information.

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1-4 SI PCF Warranty Registration

Important Note for the Installer

This *Registration Form* MUST be completed by the person undertaking installation and witnessed by the owner of the product. In the event of a Warranty Claim, the information contained within this Registration is your evidence that the product has been correctly installed. Failure to provide this evidence in the event of a warranty claim may result in the claim being rejected (*this does not affect your statutory rights as a customer*). A copy of this form must be sent to your local Medisafe authorised distributor to validate the warranty.

Tests parts one and two must be completed by the engineer commissioning the SI PCF for warranty registration purposes. Where no value is required write pass or fail with a comment if necessary.

For installation instructions and contact information for qualified personnel who can perform the installation, please contact Medisafe (see section 1-1).

Owner Name:	
Installation Address:	
Installation Date:	
Model No:	
Serial No:	
Software Version:	
Installed by:	
Witnessed by:	
Position:	

Part 1, Test Type	Acceptable Value	Result	Comments
Drains remove effluent effectively	-		
Phase sequence, Pump Rotation	-		
Support bases secure	-		
There is no evidence of interference to or from other equipment connected to the same services	-		
Hot Water Hardness (As specified by WD Manufacturer)	125 ppm (max)		
Cold Water Hardness (*As specified by WD Manufacturer*)	125 ppm (max)		

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Part 1, Test Type	Acceptable Value	Result	Comments
Hot Water Temperature	50°C (122°F) min 80°C (176°F) max		
Cold Water Temperature	5°C (41°F) min 30°C (*86°F*) max		
Hot Water Pressure	2 - 5 bar (30 – 70 psi)		
Cold Water Pressure	2 – 5 bar (30 - 70 psi)		
Part 2, Final rinse water	**Acceptable Value**	**Result**	**Comments**
Rinse water supply Type	RO, DI, Softened		
If softened, Hardness	< 125 ppm		
Water Pressure	2 – 5 bar (30 – 70 psi)		
Temperature	5°C (41°F) min 80°C (176°*F*) max		
Electrical conductivity	<30µS/cm (*RO*) <10µS/cm (*DI*) <300µS/cm (*Soft*)		

Signed	Name	Date

Comments / remedial work	

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1-5 Safety Precautions

The following *Safety Precautions* **must** be observed when operating or servicing the SI PCF. WARNING indicates the potential for personal injury and CAUTION indicates the potential for damage to equipment. For emphasis, certain *Safety Precautions* are repeated throughout the manual. **It is important to review ALL *Safety Precautions* before operating or servicing the unit**.

Warning – Breathing Hazard	
	Some cleaning solutions emit harmful vapours. Only use cleaning solutions in well-ventilated areas. Do not inhale vapours. Do not permit solution vapours to exceed maximum allowable concentrations in the work area. Ensure adequate ventilation.
Warning – Chemical Burn and/or Eye Injury Hazard	
	• Detergents are caustic and can cause adverse effects to exposed tissues. Do not get in eyes, on skin or attempt to ingest by mouth. • Read and follow precautions and instructions on detergent label and in the Material Safety Data Sheet (MSDS) prior to handling chemical, refilling chemical containers or servicing chemical injection pumps and lines. • Wear appropriate Personal Protective Equipment (PPE) whenever handling chemicals or servicing chemical injection pumps and lines. This should include plastic gloves, goggles or full-face visor, and disposable plastic apron. • Wear appropriate Personal Protective Equipment (PPE) whenever removing baskets or parts from cleaning chamber. • In case of body contact with chemical, always follow emergency procedure given in that chemical's MSDS or call emergency services. • Use a funnel when transferring any chemical from one container to another.
Warning – Burn Hazard	
	• Since temperatures within the unit can reach 92°C (198°F) during and after operation, take care when opening lid at the end of a cycle. Allow sufficient time for items to cool before handling, cleaning or maintenance procedures. • Wear appropriate Personal Protective Equipment (PPE) whenever reaching into Chamber after a thermal disinfection operation. • Under no circumstances attempt to forcibly open the lid during operation. • During reloading of instruments in the re-wash program the water temperature within the unit can reach 60°C (*140°F*), take care when opening lid and reloading the instruments.

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Warning – Explosion Hazard	
 Warning	Use only water or water-detergent mix in cleaning chamber. Never use flammable or combustible cleaning liquids. Fire or explosion could result.
Warning – Electric Shock Hazard	
	• To prevent the risk of fire or electric shock, do not expose equipment to unnecessary moisture. Always ensure that all external surfaces are dried after each cycle. • When removing a basket, allow to drain into the chamber before moving to the next cleaning station. Excess water spillage may cause electrical shock and serious injury. • Since dangerous high voltages are present inside the unit, keep the left cabinet door closed and locked at all times. • Disconnect all utilities to unit before servicing. Do not service unit unless all utilities have been properly locked out. Always follow local occupational health and safety regulations, as well as electric and plumbing codes. • Isolate electrical supply by turning the buildings disconnect switch to OFF before checking or replacing any parts. Electrical shock can cause serious injury.
Warning – Slipping Hazard	
 Warning	To avoid slippery floor conditions, immediately wipe up any spilled liquid or condensation. If spills or leaks could contain detergents or other chemicals, follow safety precautions and handling procedures set forth on detergent or chemical label and/or MSDS.
Warning – Personal Injury and/or Equipment Damage Hazard	
 Warning	• Only Medisafe or Medisafe-trained service personnel must make repairs and adjustments to this equipment. Maintenance performed by unqualified personnel or installation of unauthorized parts could cause personal injury, result in improper equipment performance, invalidate the warranty, or result in costly damage. Contact Medisafe regarding service options. • Regularly scheduled preventive maintenance, in addition to the faithful performance of routine maintenance described within the service manual, is required for safe and reliable operation of this equipment. Contact Medisafe to schedule preventive maintenance. • Failure to perform periodic service inspections of the machine could result in serious personal injury or equipment damage. • Do not wear loose fitting clothing, neckties or jewellery that could become entangled in moving parts. • Never reach into a chamber when unit is in operation. • Always use a basket to reprocess instruments within the unit.

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Caution – Possible Equipment Damage	
 **Caution**	• Never place items or person(s) on the lid. • Do not drop items into the wash tank as this could damage the ultrasonic transducers. • Do not place items directly into the wash tank or allow them to make contact with the tank sides. • Never scratch chamber walls. This can increase cavitation erosion. • Do not load basket with instruments pointing at the lid. • Do not overload the basket, maximum load of 10kg (22lbs). • Do not allow Luer pipes to protrude from the basket. • When choosing a detergent, select one with a low chloride content. Detergents with a high chloride content can be harmful to stainless steel. • Repeated fuse blows or circuit breaker trips indicate a short circuit or overload. Have a fully trained technician locate and correct situation. • Use nonabrasive cleaners when cleaning unit. Follow directions on container and rub in a back-and forth motion, in same direction as surface grain. Abrasive cleaners damage stainless steel. Cleaner rubbed in a circular motion or applied with a wire brush or steel wool will scratch and dull stainless steel. • Avoid running equipment into the Control Panel or Touch Screen to prevent damage to controls.
Caution – Chemical	
 **Caution**	• Chemical bottles must be connected to the correct port. Otherwise, wash efficiency and/or rinsing will be compromised. • Check chemical bottles are in date before use
Caution – Instrument Care	
 **Caution**	• Always confirm compatibility with the instrument manufacturer. • Always follow the manufacturer's instructions if processing instruments with special characteristics e.g. fibre optics or rigid optics. • Whether pre or post processing instruments, always follow the manufacturer's instructions. • Always attach tubing to a suitable size distal connector for instruments being cleaned or to a quick connect port if the instrument has one. • Always open box jointed instruments. • Always ensure taps are open on cannulated instruments. • Always dissemble instruments that are designed to be taken apart. • Always attach at least one Luer pipe to each basket manifold if cannulated instruments are not being cleaned, this will aid water circulation. • Visually inspect instruments after reprocessing • If any instruments became detached from the Medisafe connection tubing during processing, they should be reprocessed after ensuring that the Luer connection is firmly seated. • Do not connect more than ten instruments to the Si Robotic Basket as this will reduce the water pressure and compromise effective cleaning. • Do not connect more than six instruments to the Xi Robotic Basket as this will reduce the water pressure and compromise effective cleaning.

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1-6 Training Personnel

The owner of this equipment has the responsibility for training the users and keeping the appropriate records.



There should be regular training of all personnel concerned with the operation and maintenance of the equipment. The training schedule shall include:

- Operation of the machine (see section 2-10, 3-1, 3-4-1, 3-4-2, 3-5, 3-7, 3-8, 3-9, 3-10 and 3-11).
- Safe use of machine (see section 1-5)
- Chemical handing (see section 3-6)
- Emergency Procedures e.g. how to handle the load in case of aborted cycle (see section 3-12 and 3-13)
- Regular maintenance procedures (see section 3-15 and 3-16)
- Troubleshooting (see section 4)

The following table can be used for keeping training records.

Date	Name	Title of Training	Signature of Trainer

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1-7 Compliance

The SI PCF has been designed to perform within the guidelines recommended by international decontamination Standards. A stringent quality control program ensures that every unit is manufactured and tested under controlled conditions to ensure that all products perform identically.

Commitment to RoHS and WEEE Requirements

Medisafe is committed to meeting or exceeding the requirements of the RoHS CE Marking directive (2011/65/EC) (*Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment*).

Medisafe is also committed to meeting all requirements of the WEEE directive (2012/16/EU) (*Waste Electrical and Electronic Equipment*).

WEEE and Battery Disposal



Waste electrical and electronic products must not be placed in the general refuse but collected separately. Separately collected WEEE products are to be sent to an approved recycling or disposal operators for safe recycling, recovery or reuse.

Medisafe UK Ltd is registered with the "B2B Compliance" Scheme. B2B Compliance on behalf of Medisafe, takes on the legal responsibilities of the reporting, collection and treatment of, all WEEE that Medisafe UK Ltd is obligated for - and ensures that the appropriate recycling targets are met. .For independent confirmation of Medisafe UK Ltd status can call B2B Compliance on 0845 680 9829 or visit www.b2bcompliance.org.uk.

To fulfil our B2B WEEE take-back obligations, B2B Compliance will undertake the appropriate collection and disposal of obligated products of Medisafe UK Ltd.

If a Medisafe Product has reached its end of life, please contact B2B Compliance collections team on 0845 680 9829 or collections@b2bcompliance.org.uk to arrange collection and treatment.

Otherwise, please recycle where facilities exist. Check with your Local Authority or Distributor for recycling advice.

Declaration of CE Conformity

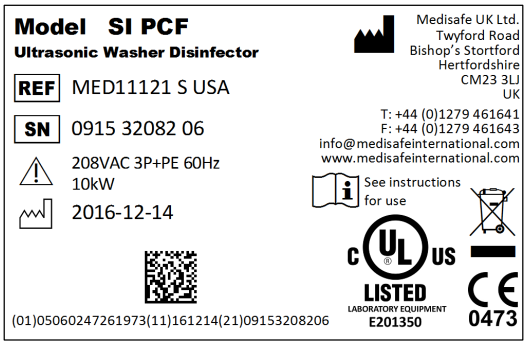
The SI PCF is designed and manufactured in the UK by Medisafe UK Ltd. and complies with the essential requirements of the Medical Devices Directive 93/42/ EEC (where appropriate). All products are CE compliant and are registered as Class IIb devices through the application of rule 15 within Annex IX of the Medical Devices Directive 93/42/ EEC.

CE 0473

1-8 Product Identity

The model and serial number label are located on the inside panel of the chemical compartment.

Example of serial number label:



The software version is shown on the front panel display when the unit is first switched on.

1-9 Manufacturer Information

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Bishop's Stortford
Hertfordshire CM23 3LJ
England

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Fax: +44 (0) 1279 461643

E-mail info@medisafeinternational.com

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1-10 Intended Use

The Medisafe Sonic Irrigator PCF is a High-Pressure, Ultrasonic, Flushing Washer Disinfector for the medical market. It is intended to provide cleaning and low-level thermal disinfection of surgical instruments, including complex lumen/cannulated devices, such as robotic surgical instruments.

This is achieved through a combination of ultrasonics and a flushing action to remove debris from both the outside and inside (hollow instruments only) of items placed within the wash tank.

Note: *The SI PCF Washer has been validated for the cleaning and low-level thermal disinfection of surgical instruments and devices, including complex lumen/cannulated surgical instruments and devices such as robotic surgical instruments. Testing to validate cleaning and disinfection efficacy for a specific instrument is the responsibility of the instrument manufacturer. Consult the instrument manufacturer's instructions regarding cleaning and disinfection.*

Note*: If the equipment is used in a manner not specified by the manufacturer, the effectiveness and protection provided by the equipment may be impaired.*

1-11 Performance

Washing is achieved using a unique, specially formulated and clinically validated decontamination fluid (EDA+ or 3E-Zyme®), which is introduced into the machine during the cycle.

This machine has a class AB weir incorporated into the tank design to comply with most water authority regulations.

A stringent quality control program ensures that every unit is manufactured and tested under controlled conditions to ensure that all units perform identically.

1-12 Cycle Record

During the cycle, parameters are reported at key stages of the process as an ASCII serial data steam via the RS232 port. This can then be logged either on a printer (see section 2-6-2 and 2-10) or on a PC (see section 2-11).

The on-board cycle log can be download via a USB flash drive (see section 3-14).

1-13 Cycle Failure

The SI PCF is designed to perform within strict and carefully controlled tolerances. If an unexpected condition occurs during a wash cycle, the system will abort the cycle and report an error, see section 3-12 for more details.

1-14 Concept Summary

The SI PCF has the ability to wash up to twenty instruments.

There are six pre-defined user selectable programs with *Pre-wash, Wash, Rinse and Disinfection* stages tailored to clean a multitude of different loads.

The program parameters can also be changed within specified limits by an engineer during installation or service.

The SI PCF has an interlock system that prevents a cycle being started with the lid up.

Load Instruments

The user loads the instruments into the basket, places it into the machine and presses the touch screen to lock the lid.

Prewash

The cycle begins with a cold-water fill to the mid-level; the spray and irrigation valves are opened alternately with the pump running for the specified time. The water is then drained.

Wash

At the main wash stage, the tank is filled to the top level with a cold and hot water mix to attain a set temperature. Detergent is also added at this point. The pump is pulsed on and off, to even out the temperature, fill all cavities, and to mix the chemical thoroughly. Once the tank is full the ultrasonics degas the tank by pulsing on and off to remove any dissolved air. A period of irrigation follows in order to flush any material from inside the instruments. The ultrasonics then alternate with irrigation for the set period of time. The water is then drained.

Rinse

The tank is filled to the mid-level; and heated to the set point. The machine then alternates between the spray and irrigation for the pre-defined time. The water is then drained.

For some programs this stage is repeated.

Disinfection

If disinfection is required, the machine is then filled to the minimum level with hot water and heated up to the pre-set temperature (*91°C / 196°F*) for 3 minutes. The water is then drained.

Cooling

To disperse the vapour in the chamber, the machine opens the drain valve and fills with cold water for several minutes.

Unload

The user touches the screen to acknowledge the end of the cycle, which unlocks the lid and allows the basket to be removed.

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1-15 General Safety

This equipment is not suitable for use with flammable materials or gases, and should not be used for any purpose other than that defined by the manufacturer as a high pressure sonic irrigation washer disinfector.

1-15-1 Chemicals

Since this equipment contains a high power heating device, the use of substances or chemicals, other than those specified by Medisafe, may be hazardous. Therefore, always read the manufacturer's chemical safety information before use and never use toxic, corrosive, or flammable liquids.

Using non-approved chemicals may cause damage to your machine and could void your warranty and guarantee of performance.

1-15-2 Decontamination

In the event that this product needs to be returned to the supplier, it **MUST** first be fully decontaminated before transporting. The *Decontamination Certificate* (see *section 3-16-4)* must also be completed and displayed in a prominent position when returned with the product.

1-15-3 Heat



Since temperatures within the unit can reach 92°C (*198°F*) during and after operation, take care when opening lid at the end of a cycle. Allow sufficient time for items to cool before handling.

1-16 Electrical Safety



Dangerously high voltages are present inside the machine. Do not remove any panelling, or attempt to gain access to the inside of the machine, keep the left cabinet door closed and locked at all times. Refer servicing and repair to qualified personnel only.

It is expected that any person servicing or repairing the SI PCF will be a competent engineer with an understanding of electrical safety, and knowledge of three phase power. It is also crucial to understand the risks of working with electrical appliances that use water. Caution must be taken at all times to ensure a safe environment.

To avoid risk of electric shock, this equipment must only be connected by a fully qualified electrician and to a mains supply with a protective earth.

To prevent the risk of fire or electric shock, do not expose equipment to unnecessary moisture. Always ensure that all external surfaces are dried after each cycle.

The key must only be available to a Responsible Body.

1-17 Service Contracts & Warranty

Your SI PCF is supplied with a conditional 12-month parts warranty, which is effective from the date of installation.

Medisafe (UK) offers a range of Service Contracts and Extended Warranty Packages to provide continuous maintenance and breakdown cover once the original warranty has expired.



Hazards can result from unauthorized modification of the machine. No modification of this equipment is allowed. Any modification to this equipment will void the warranty.

1-18 Validation Option

Medisafe (UK) can also provide a *Validation Service* to address the essential local regulation requirement. The work will be carried out by authorised Medisafe engineers to ensure your SI PCF continues to conform to local and national guidelines.

For more information on service or any of these options, please contact Medisafe or your regional distributor.

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2. Installation

User Manual



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2-1 Transport and Storage

The permissible environmental conditions for transport and storage of a new or used SI PCF must adhere to the environmental conditions for altitude, temperature and relative humidity.

2-1-1 Marking

The markings on the packing crate stipulate the machine's storage conditions and handling information.



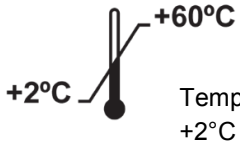
Keep dry



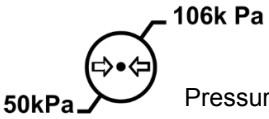
This way up'

2-1-2 Environmental

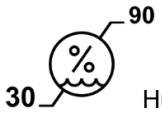
This equipment can be transported and stored safely, in the following environment:



Temperature Range:
+2°C to +60°C



Pressure range
50kPa to 106kPa



Humidity range
30% to 90% RH
(*Non-condensing*)

2-2 Lifting

Lifting and manoeuvring the SI PCF in its packing crate must be done using a pallet truck or forklift truck able to take a load in excess of 250kg.

2-3 Unpacking

Retain all packaging materials, if for any reason the unit needs to be returned to the supplier, then the original crate and packaging materials should be re-used.

Carefully unpack your SI PCF by removing the top panel first. The front panel of the packing crate is designed to be used as a ramp to aid in getting the machine off the crate. Once off its base and unpacked, it can then be moved across a level floor on its castors.

This equipment is not intended to be routinely moved, wheels are provided only to facilitate installation and servicing by a service engineer.

2-4 Returns

Before transporting a used SI PCF, it is vital that all fluid is thoroughly drained from the machine. If returning the unit to your supplier or Medisafe UK Ltd it **MUST** first be fully decontaminated before transporting.

The *Decontamination Certificate* must also be completed and displayed in a prominent position when returned with the product (see section 3-16-4).

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2-5 Enclosures Check List

The unit is supplied with a range of accessories to enable easy installation and efficient operation. Ensure that all these items are present before attempting to install or operate the unit. Complete the following Enclosures Check List and notify your supplier of any shortages or damaged components as soon as possible.

The SI PCF comes with one basket (standard), the type of basket and typical accessories supplied for each SI PCF variant are shown in the tables below. Replacement items can be ordered, see below list for details.

Note: *The below depends on your machine type, please refer to what has been specified on your order.*

2-5-1 Standard PCF S

Standard PCF S Machine Part Numbers
MED11100 MED11110 MED11122 MED11121 MED11121UL MED11120 MED11123

Quantity	Part No.	Description	Checklist ✓
1	MED8601	Standard Basket	
1	MED1014x	Thermal printer (For country specific version see 2-6-2)	
1	MP21424	Standard PCF Accessory Kit	

2-5-2 Robotic PCF S

Robotic PCF S Machine Part Numbers
MED11130 MED11131 MED11132 MED11133

Quantity	Part No.	Description	Checklist ✓
1	MP18990	Si Robotic Basket	
1	MED1014x	Thermal printer (For country specific version see 2-6-2)	
1	MP21430	Standard PCF Accessory Kit	

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2-5-3 Standard Accessory Kit (MP21424)

Quantity	Part No.	Description	Checklist ✓
1	MED8557	Detergent Pick Up Assembly	
2	MED8564	Instrument Organiser	
1	MET129	Water Heater Spanner	
3	HOSE007	Water Inlet Stainless Steel Hose – Red Indicator	
1	HOSE010	Overflow Hose clear PVC reinforced 50mm I/D	
1	MED3020	Drain Hose 2.5m	
1	MED8558	Jubilee Clip 50-70mm	
1	MED1017	Printer Serial Data Cable DB9 Male to Female Straight through	
20	M20003	Straight Hose & Luer (*240mm*)	
2	MED1246.5.0	Distal Connector box of 10 mixed	
1	SWITCH004	Lid Lock Override Key	
1	SWITCH006	Electrical Cabinet Door Key	

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2-6 Additional Accessories

A range of Medisafe approved accessories including replacement parts and consumables, may be purchased from your supplier. The use of these compatible Medisafe approved products will ensure the continued and reliable operation of your machine.

Note: *Using non-approved chemicals, accessories, and replacement parts may cause damage to the machine and could void both your warranty and your guarantee of performance.*

2-6-1 Baskets

Part No.	Description
MED8601	Standard Basket
MP21833	Si Robotic Basket – Including Hose & Luers
MP21104	Xi Robotic Basket – Including Hose & Luers

2-6-2 Printer

Part No.	Description
MED1014B	Thermal printer including power supply, paper and USA plug
MED1014C	Thermal printer including power supply, paper and European plug
MED1014F	Thermal printer including power supply, paper and Australian plug
MED1014G	Thermal printer including power supply, paper and UK plug
MED1014M	Thermal printer including power supply, paper and South Africa plug
MED1014.1	Thermal printer paper (1 x 25m roll)

2-6-3 Approved Chemicals

Part No.	Description
MED7701	EDA+ 10L Bottle
MED8037	3Ezyme 10L Bottle

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2-7 Installation Requirements

Machine Position	
Site	Level floor and at least 150mm (6in) away from surrounding walls for ventilation (see section 2-9 for dimensions and clearances). There should be enough free space provided near the machine for safe and efficient maintenance in the future.
Ambient Lighting	500 Lux minimum room lighting

Mains Supply

Cables shall be rated for maximum current of the equipment, meeting the requirements of IEC 60227 or IEC 60245. The SI PCF is supplied with 1.8m (6ft) of screened power cable.

USA Model MED11121UL (only)
Power cable must be wired using a plug and socket.

For all Other Models
Power cable must be permanently installed by wiring directly into a switched wall panel. Under no circumstances should it be wired using a plug and socket.

Conductor Colour Coding

380-415V 3~ Models	200-230V 3~ Models	Single Phase Models
MED11120, MED11132 MED11123, MED11133	MED11122, MED11130, MED11121, MED11121UL, MED11131	MED11100 MED11110
Live 1: Brown Live 2: Black Live 3: Grey Neutral: Blue Earth: Green/Yellow	Live 1: Black Live 2: Black Live 3: Black Earth: Green/Yellow	Live: Brown Neutral: Blue Earth: Green/Yellow

The cable must be protected with a 3phase Residual Current Device (RCD) and Miniature Current Breaker (MCB). The RCD and MCB shall only be used to supply this equipment.

The disconnect switch must:

- Isolate all poles simultaneously
- Be lockable in the off position
- Comply with standards IEC/EN 61058-1 and IEC 60447 (creepage and clearances for 4kV transient voltage)
- Be situated close to the equipment for ease of accessibility, and in a position where the operator can turn it off in an emergency without leaning over the machine.
- Marked as the disconnection device for this equipment
- Disconnect device off-position should be clearly marked (O) and the on-position should be marked (I)



To avoid risk of electric shock, this equipment must only be connected by a fully qualified electrician and to a mains supply with a protective earth.

Supply Voltage	See Section 5-1
Position of supply	<1.8m from power cable exit. (*Rear lower left of unit*) If >1.8m please advise
Electro Magnetic Compatibility (EMC)	Medical electrical equipment needs special precautions regarding EMC and shall be installed according to the EMC information provided in the Appendix

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Water Supply	
Water Hardness ($CaCO_3$), Quality & Purity	<125ppm with water quality and purity tested and approved for Disinfection as per EN ISO15883-1:2009 (*item 6.4.1 and 6.4.2*)
Pressure	2 – 5 bar (*30 – 70 psi*) for all water supplies
Flow Rate	12 – 30 l/m (*3 – 8 US gal/m*)
Temperature:	Cold Water 5 – 30°C (41 – *86°F*)
	Hot Water 50 – 80°C (*122 – 140°F*)
	RO Water 5 – 80°C (*41 – 140°F*)
Connection	1x shut valve with ¾in BSP male thread for hose tail (*Hot water*)
	1x shut valve with ¾in BSP male thread for hose tail (*Cold water*)
	1x shut valve with ¾in BSP male thread for hose tail (*RO water*)
Position	Within 1m (*40in*) of proposed installation

Drainage System	
Type	Pumped
Drain Connection	19mm (¾*in*) hose tail outlet from machine Drain must fitted with suitable traps No Loops in the drain hose, cut the hose to required length
Position	Drain entry and the entire drain hose must less than 500mm (20in) above ground level sited at the rear middle of the machine Drain gulley pipe ID must be at least 100mm (*4in*) to conform with ISO15883
Capacity	> 30ltr/m (*8 US gal/m*)
Overflow	48.3mm (*1.9in*) OD outlet from machine – into the same gulley as the drain The overflow hose connection to the main drainage system must not be airtight
Effluent Temp	95°C (203°*F*)

Printer	
Dimensions	W:230mm (*9in*) x D:280mm (*11in*)
Position	Above or to the side and at least within 1m (*40in*) from the machine, where it will not be affected by spillage, leakage or condensation from the equipment
Supply Connection	90 - 230Vac 50/60Hz (*single phase*)
Position of printer power supply	Within 1m (*40in*) from the machine, where it will not be affected by spillage, leakage or condensation from the equipment

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2-8 Connecting the SI PCF

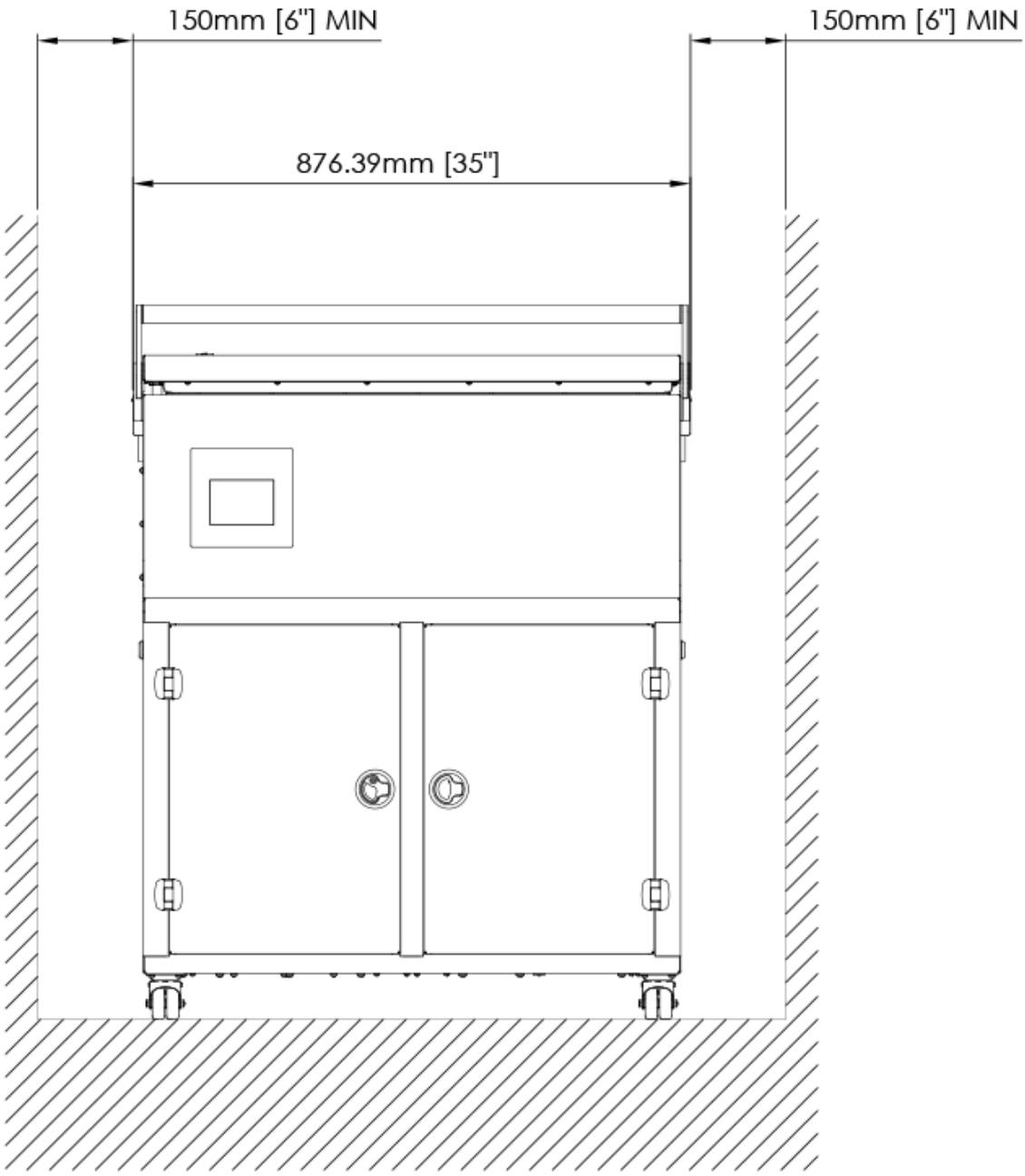
***Note**: Medisafe (UK) Ltd. recommends this product be installed & commissioned by Medisafe (UK) Ltd (or suitably qualified personnel).*

1. Unpack the machine and position it within 1m of the mains power supply, drain, and mains water supplies.
2. Remove any protective packaging and loose items from inside the wash tank.
3. Identify the grey plastic, convoluted drain hose supplied with the unit. Connect the small end of the hose to the ¾" drain hose tail at the lower rear of the unit.
4. Secure the drain hose to the machine using the supplied jubilee clip.
5. Connect the large end of the drain hose to the site drain, cutting to length as required.
6. Identify a stainless steel water hose supplied with the unit. Ensure the seal in the hose end is in place and screw the elbow connector onto the cold-water inlet at the lower rear of the unit.
7. Connect the straight end of the hose to a suitable cold water supply.
8. Repeat for the hot water supply.
9. Repeat for the RO water if this has been made available on site.
10. Turn on the water supplies and check for leaks.
11. Connect the machine to the electrical mains supply (see section 2-7).
12. Carry out the commissioning procedure defined within the SI PCF Service Manual.

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2-9 Installation Drawings

2-9-1 Front View

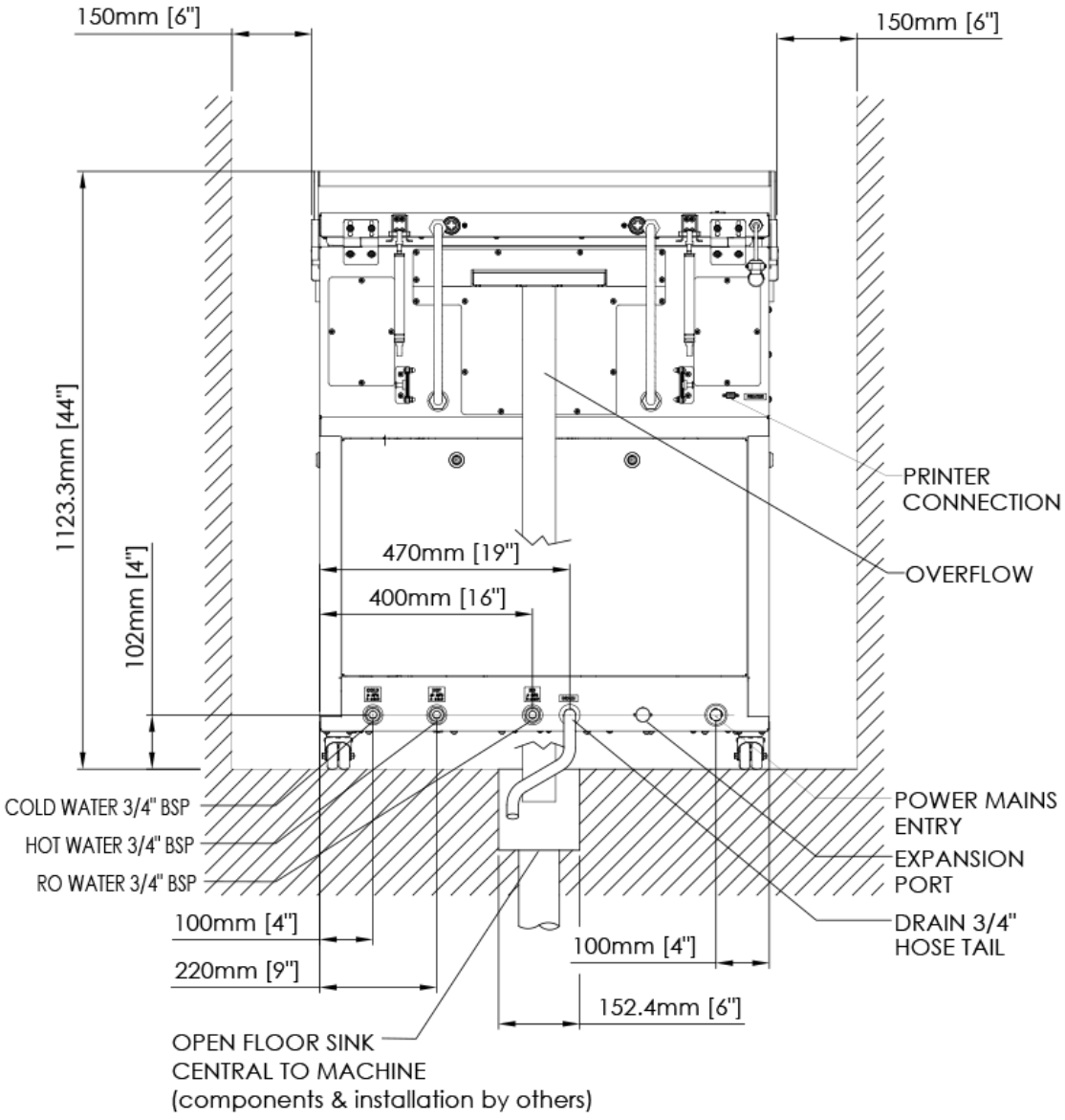


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2-9-2 Rear View Showing Services

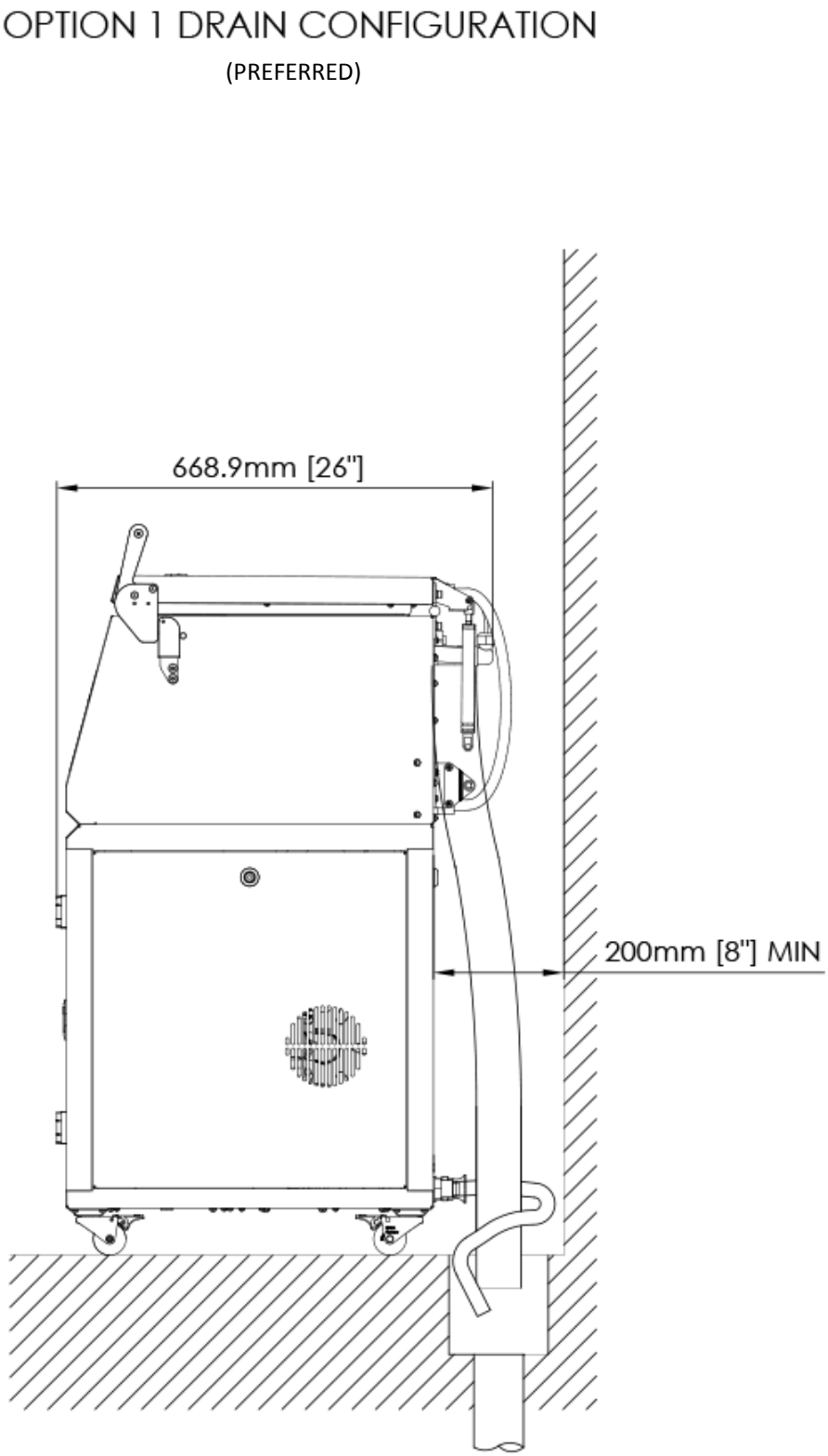
OPTION 1 DRAIN CONFIGURATION

(PREFERRED)



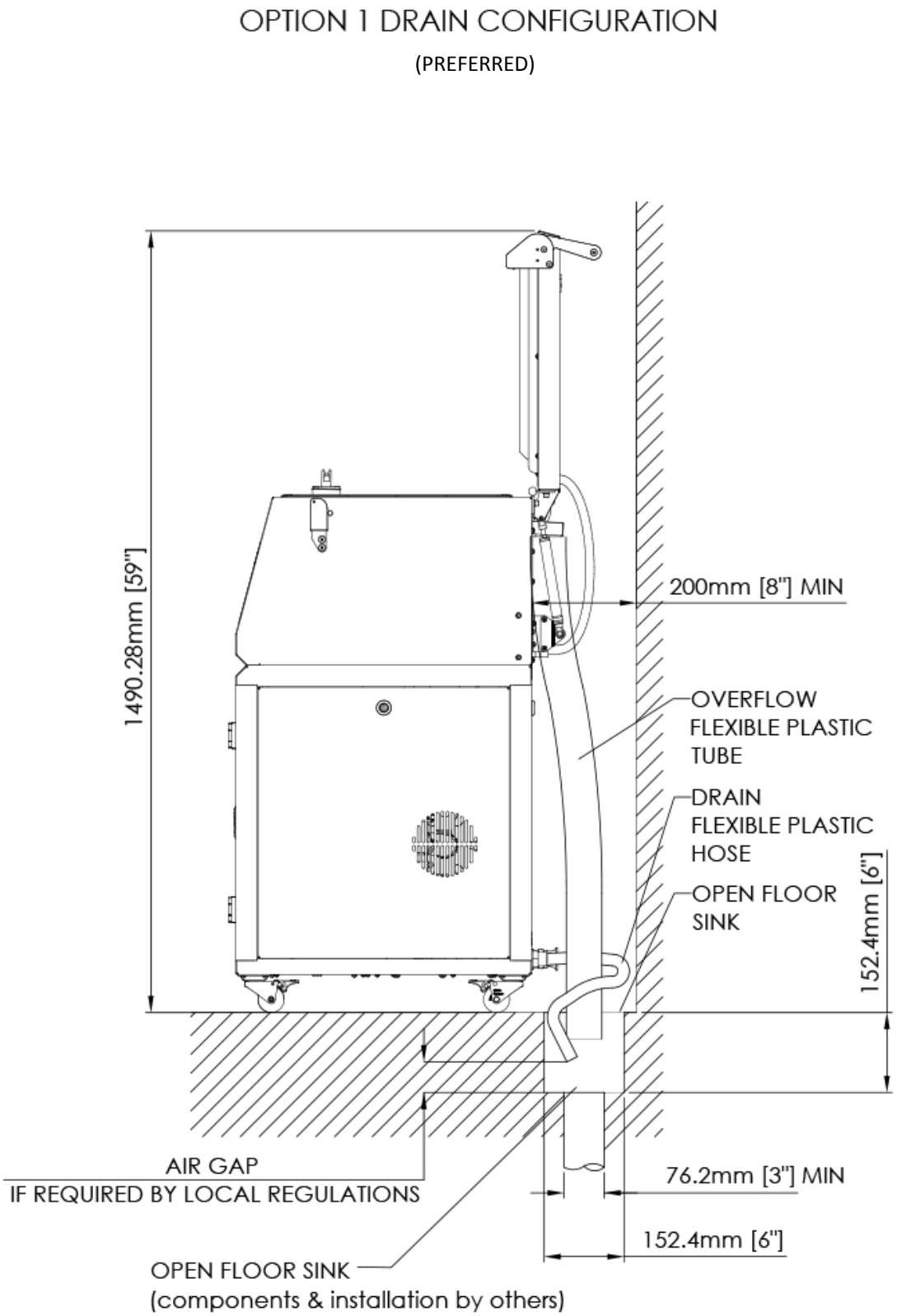
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2-9-3 Right Side View Lid Closed



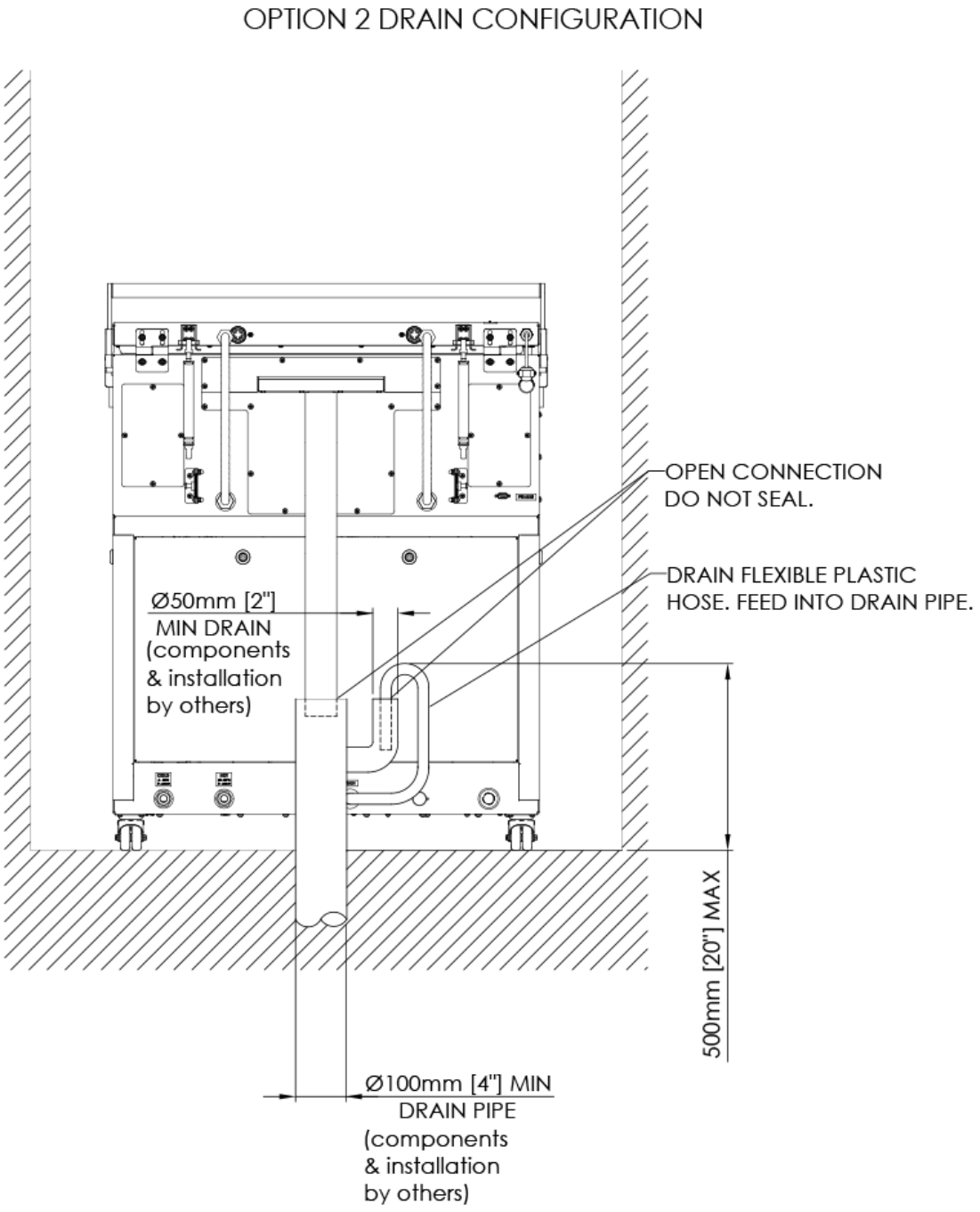
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2-9-4 Right Side View Lid Open



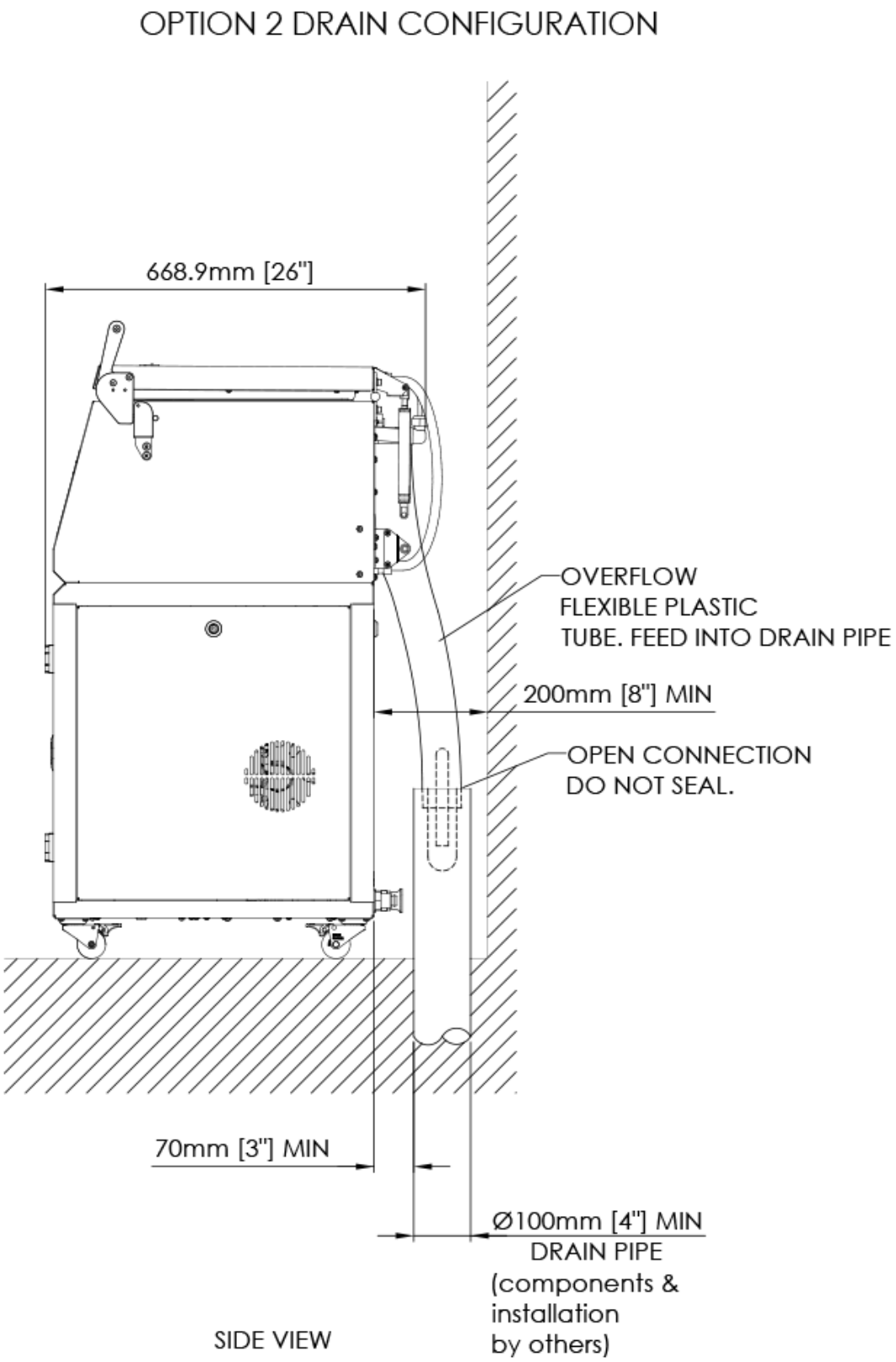
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2-9-5 Rear View Alternative Drain Connection



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2-9-6 Right Side View Alternative Drain Connection

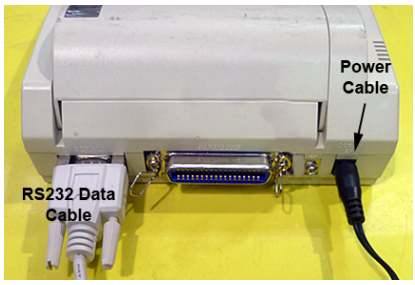


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Connecting to the RS232 Port

2-10 Thermal Printer

1. Identify the RS232 Data Cable supplied, and plug the correct end into the socket located at the lower right hand side at the rear of the unit.
2. Connect the other end of the RS232 data cable into the connector on the rear of the thermal printer.



3. Plug the printer power supply cable into the rear socket of the printer, and connect the mains power cable into a suitable mains supply outlet, then switch on the printer.

Note: *When the printer paper shows red stripes, this indicates the paper is about to runout, and must be replaced. Once the paper has been changed press the online button, the green led will illuminate and printing will resume.*

The printer can store more than 10 cycles in memory when the paper has run out. If the power is not removed when changing the paper, the stored cycles will be printed after the paper has been changed.

2-11 Data Logging PC

Wash cycle data is sent from the serial port to a printer, however, as an alternative data may be sent to a logging device or computer via the serial port using a serial data logging program of your choice.

Note: *Only IEC/EN60950-1 certified equipment should be connected.*

Connect a suitable RS232 Null modem cable between the data port on the rear panel and a COM port on your PC. Run your serial data logging program, ensuring it is configured in the following manner:

- Baud Rate – *19200bps*
- Data Bits – *8*
- Parity – *none*
- Stop Bits – *1*
- Flow Control – *none*
- Handshaking – *off*
- Buffer size > 500
- Append line feed to cartridge return

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3. Operation

User Manual



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3-1 Menu

The SI PCF is controlled using a touch screen menu. The scroll ⟨ ⟩ arrows are used to cycle through the menu pages.



Menu Functions

	**Start**	Start the automated cycle sequence
	**Wash Program**	Set the selected wash program
	**Chemical 1**	Prime chemical pump 1
	**Chemical 2**	Prime chemical pump 2
	**Clock**	Set the time and date
	**Data Logging**	Download logged cycle data onto a USB Flash Drive
	**Supervisor Menu**	View and Edit Program Parameters
	**Information**	Serial number, cycle count and software version

***Note**: When the SI PCF has been idle for 10 minutes the screensaver will start, touch the display to exit.*

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3-2 Power Up

Turn on the power to the machine using the isolation switch on the wall.

When the machine is turned on, it conducts a Power-On Self-Test (POST) to ensure the machine is ready for use.

This process takes a few seconds to configure the machine and check the basic operation of the control system, memory and sensor interfaces.



3-3 Authorisation code

This is an optional feature, which can be enabled by a service engineer. When enabled, the user is required to enter a four-digit PIN number to unlock the user control panel. A valid PIN entry identifies the user and their operator number is shown on the cycle printout. There are 80 operator numbers each with a unique PIN number, see Appendix 6-8 for details.



3-4 Setting Time & Date

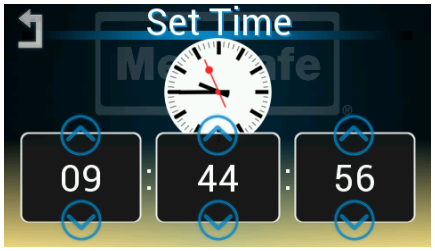
The time and date must be set correctly to ensure the accuracy of the cycle records, and **MUST** be periodically checked by the user. There is no automatic provision for daylight saving hours. While the machine is off, a backup battery maintains the time and date.

3-4-1 Set Time

1. Use the scroll ⟨ ⟩ arrows to cycle through to the clock icon



2. Press ***Set Time***
3. Use the ⌃ ⌄ arrows to adjust the Hours, Minutes and Seconds



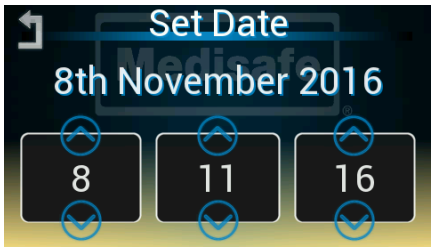
4. Press ↰ to return to the main menu

3-4-2 Set Date

1. Use the scroll ⟨ ⟩ arrows to cycle through to the clock icon



2. Press ***Set Date***
3. Use the ⌃ ⌄ arrows to adjust the Date, Month and Year



4. **Press** ↰ to return to the main menu

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3-5 Selecting a program

The SI PCF is installed with six pre-programmed wash cycles during manufacture. When starting a cycle, the machine will use the currently selected program. To change the selected program, prior to initiating a cycle:

1. Use the scroll ⟨ ⟩ arrows to cycle through to the Wash program icon

2. Change the selected program with the **+** and **–** buttons



3-5-1 Standard Program Parameters

Parameters	Units	Program Number					
		Standard Wash Cycle	Re-Wash Cycle	Heavy Wash Cycle	Standard Disinfect Cycle	Heavy Disinfect Cycle	Intense Disinfect Cycle
		P1	P2	P3	P4	P5	P6
Pre-wash Time	Minutes	3	0	3	3	3	3
Wash Chemical 1	ml	360	360	360	360	360	360
Wash Temperature	ºC	40	40	40	40	40	40
Degas	Minutes	3	3	3	3	3	3
Wash Time	Minutes	5	10	20	5	10	20
Rinse Temperature	ºC	60	60	60	60	60	60
Rinse Time	Minutes	2	2	2	2	2	2
Disinfection Temperature	ºC	0	0	0	90	90	90
Disinfection Time	Minutes	1	0	1	1	1	1
Typical Total Cycle Time	Minutes	40	10*	50	55	60	70

*Per Re-wash, see section 3-11-10

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3-5-2 Robotic Program Parameters

Parameters	Units	Program Number					
		Standard Wash Cycle	Re-Wash Cycle	da Vinci Wash Cycle	Standard Disinfect Cycle	Heavy Disinfect Cycle	da Vinci Disinfect Cycle
		P1	P2	P3	P4	P5	P6
Pre-wash Time	Minutes	3	0	3	3	3	3
Wash Chemical 1	ml	360	360	360	360	360	360
Wash Temperature	ºC	40	40	40	40	40	40
Degas	Minutes	3	3	3	3	3	3
Wash Time	Minutes	5	10	20	5	10	20
Rinse Temperature	ºC	60	60	60	60	60	60
Rinse Time	Minutes	2	2	2	2	2	2
Disinfection Temperature	ºC	0	0	0	91	91	91
Disinfection Time	Minutes	1	0	1	3	3	3
Typical Total Cycle Time	Minutes	45	10*	60	65	70	80

*Per Re-wash, see section 3-11-10

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3-6 Chemical Handling



Warning

Users must read this section prior to any chemical handling.

- Detergents are caustic and can cause adverse effects to exposed tissues. Do not get in eyes, on skin or attempt to ingest by mouth
- Read and follow precautions and instructions on detergent label and in the Material Safety Data Sheet (MSDS) prior to handling chemical, refilling chemical containers or servicing chemical injection pumps and lines
- Wear appropriate Personal Protective Equipment (PPE) whenever handling chemicals or servicing chemical injection pumps and lines. This should include plastic gloves, goggles or full-face visor, and disposable plastic apron
- Wear appropriate PPE whenever removing baskets or parts from cleaning chamber
- In case of body contact with chemical, always follow emergency procedure given in that chemical's MSDS or call emergency services
- Guidelines to be followed in case of emergency caused by chemical contact should be prominently displayed near the machine
- Use a funnel when transferring any chemical from one container to another
- Optional lubricants and any other chemical have to be validated and proven compatible with the PCF by its manufacturer

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3-6-1 Chemical Consumption

A 10-litre detergent bottle will give approximately 27 washes.

3-6-2 Chemical Replacement

The right hand door allows access to the chemical storage compartment, where the pickup connections are located.

The machine uses a float switch in the bottle to detect when there is insufficient chemical to run a cycle, prompting the user to replace the bottle. This assembly is inserted into the chemical bottle until the base of the float switch rests on the bottom.

Chemical 1 - Detergent

The Detergent sensor plug and pickup tube connect to the left hand socket and port.

Chemical 2 - Lubricant

The Lubricant sensor plug and pickup tube connect to the right hand socket and port.

Caution

Chemical bottles **must** be connected to the correct port. Otherwise, wash efficiency and/or rinsing will be compromised.

3-6-3 Disposal of chemical containers

For the correct disposal of chemical containers, please refer to the manufacturer's instructions and local regulations.

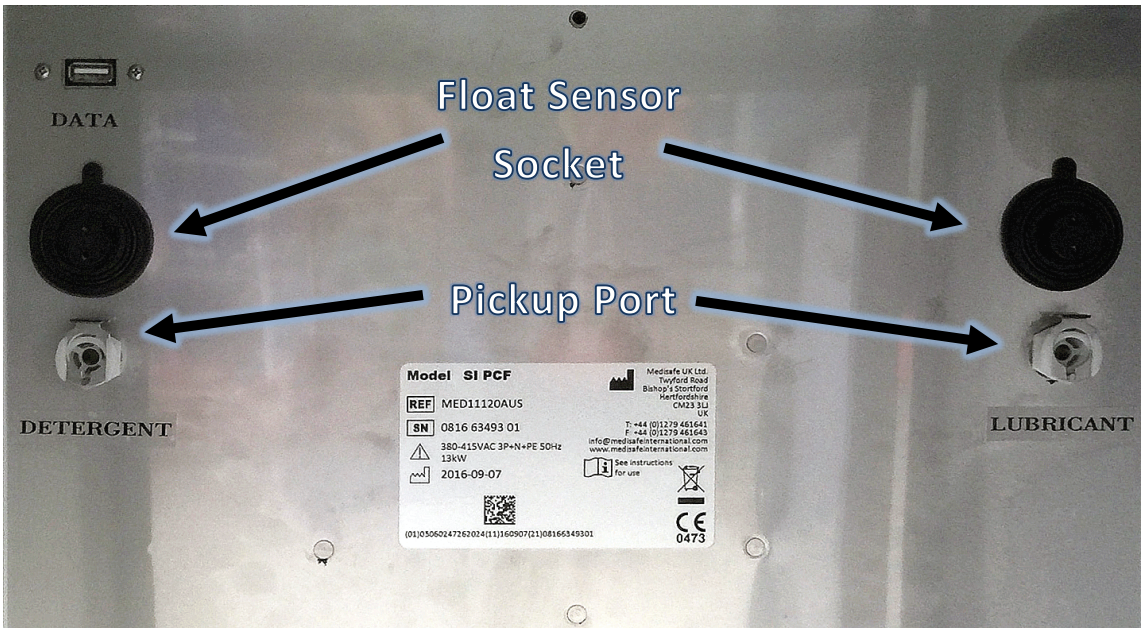
3-6-4 Chemical Priming

When changing an empty detergent bottle, to prevent unnecessary cycle errors the system should be primed to ensure chemical fills the entire hose, from the supply to the tank.

1. Remove the empty container
2. Place the new container in the chemical compartment
3. Place the intake hose in the container, ensure any spillages are cleaned up
4. Use the scroll ⟨ ⟩ arrows to cycle through to the relevant *Prime Chemical Pump* icon ⚗1 or ⚗2



5. Press ***Prime Chem. Pump*** to start the chemical pump
6. When chemical can be seen entering the tank, press ***Prime Chem. Pump*** again to stop the pump



PCF Chemical Cabinet Connections

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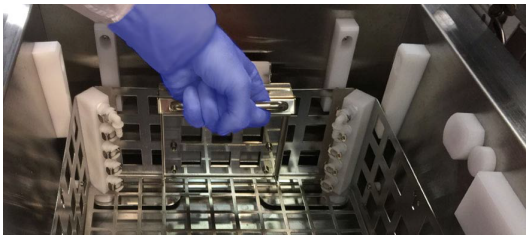
3-7 Baskets

The SI PCF basket is used to hold the instruments during the wash process. It is easily placed into and removed from the machine, utilising a drop and lock retaining feature. This prevents the basket from moving during operation, and ensures maximum irrigation pressure.

The standard basket is equipped with twenty ports for irrigation of up to 20 hollow instruments. These are divided equally between each corner to prevent crossovers or shadowing of instruments.

3-7-1 Basket Removal

To remove the basket, open the lid of the machine and pull the handles up to disingage the latch, then lift from the machine.



3-7-2 Basket Insertion

Lower the basket into the machine pushing down on the handles.



Push down until the side latches engage on both sides of the tank as shown.



3-8 Instruments

The basket ports are divided equally between each corner to prevent crossovers or shadowing of instruments.

When selecting Instruments for the SI PCF:

- **Always** confirm compatibility with the instrument manufacturer.
- **Always** follow the manufacturer's instructions if processing instruments with special characteristics e.g. fibre optics or rigid optics.
- Whether pre or post processing instruments, **Always** follow the manufacturer's instructions.

3-8-1 Instrument Loading Precautions

When loading instruments into a basket observe the following precautions:

- **Do not** drop items into the wash tank as this could damage the ultrasonic transducers.
- **Do not** place items directly into the wash tank or allow them to make contact with the tank sides
- **Do not** load basket with instruments pointing at the lid
- **Do not** overload the basket, maximum load of 10kg (22*lbs*)
- **Do not** allow Luer pipes to protrude from the basket
- **Do not** connect more than ten instruments to the Si Robotic Basket as this will reduce the water pressure and compromise effective cleaning.
- **Do not** connect more than six instruments to the Xi Robotic Basket as this will reduce the water pressure and compromise effective cleaning.
- **Always** attach tubing to a suitable size distal connector for instruments being cleaned or to a quick connect port if the instrument has one
- **Always** open box jointed instruments
- **Always** ensure taps are open on cannulated instruments
- **Always** dissemble instruments that are designed to be taken apart
- **Always** attach at least 1 Luer pipe to each basket manifold if cannulated instruments are not being cleaned, this will aid water circulation

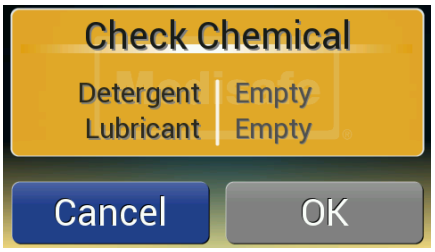
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3-9 Starting a Standard Instrument Cycle

1. Confirm program selection (to change see section 3-5)

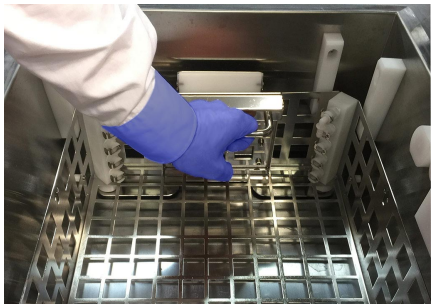


2. Press ***Load Instruments***
3. If there is insufficient chemical to complete the cycle a warning will be shown:



   Replace chemical(s) (see section 3-6-1) and press OK to continue, or press ***Cancel*** to abort the cycle start

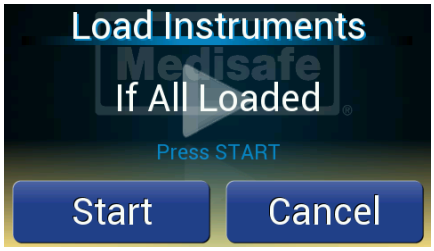
4. Open the lid
5. Remove the basket from the machine
6. Load instruments into the basket
7. Insert the basket into the machine
8. Lower the basket into the machine, pushing down on the handles.



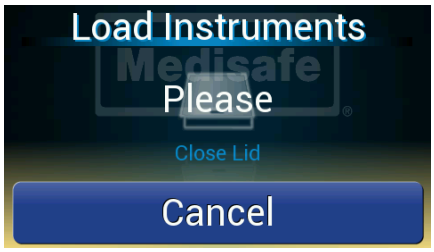
9. Ensure the basket is locked into position by pushing it down until the side latches engage on both sides of the tank



10. Press ***Start*** to confirm instrument loading complete



11. Close the lid and firmly push down on the handle to engage the lock, and the cycle will begin



Note: *The lid has a tight seal and an interlock, which requires the lid to be fully down and latched before the cycle can start (the red interlock switch should be in the locked position).*

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3-10 Starting a Robotic Instrument Cycle

3-10-1 Manual Pre Wash

Always follow the manufacturer's instructions.

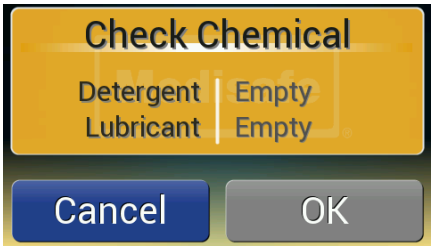
1. Scrub the outside of the entire instrument using running water with a clean, soft, nylon bristled brush, paying special attention to the instrument tip. Move the instrument wrist through its full range of motion while scrubbing
2. Thoroughly rinse the entire instrument especially the instrument tip

3-10-2 Instrument Loading

1. Confirm program selection (to change see section 3-5)



2. Press ***Load Instruments***
3. If there is insufficient chemical to complete the cycle a warning will be shown:



Replace chemical(s) (see section 3-6-1) and press ***OK*** to continue, or press ***Cancel*** to abort the cycle start

4. Open the lid
5. Remove the basket from the machine
6. Load instruments into the Si or Xi Robotic Basket depending the type to be cleaned:

   Connect the instrument to the Medisafe Robotic basket ONLY. This basket is clearly marked as 'Sonic Irrigator'.

7. Attach the 'Slip Luer' end of the Medisafe connection tubing to the main flush port of each instrument by firmly pushing to ensure that the Luer is seated.

Si Instrument Connection

The main flush port is designated by a dark ring and the number 1.



Xi Instrument Connection



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9. Lower the basket into the machine pushing down on the handles.



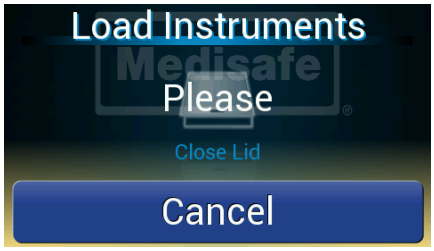
10. Ensure the basket is locked into position by pushing it down until the side latches engage on both sides of the tank



11. Press ***Start*** to confirm instrument loading complete



12. Close the lid and firmly push down on the handle to engage the lock, and the cycle will begin



Note: *The lid has a tight seal and an interlock, which requires the lid to be fully down and latched before the cycle can start (the red interlock switch should be in the locked position).*

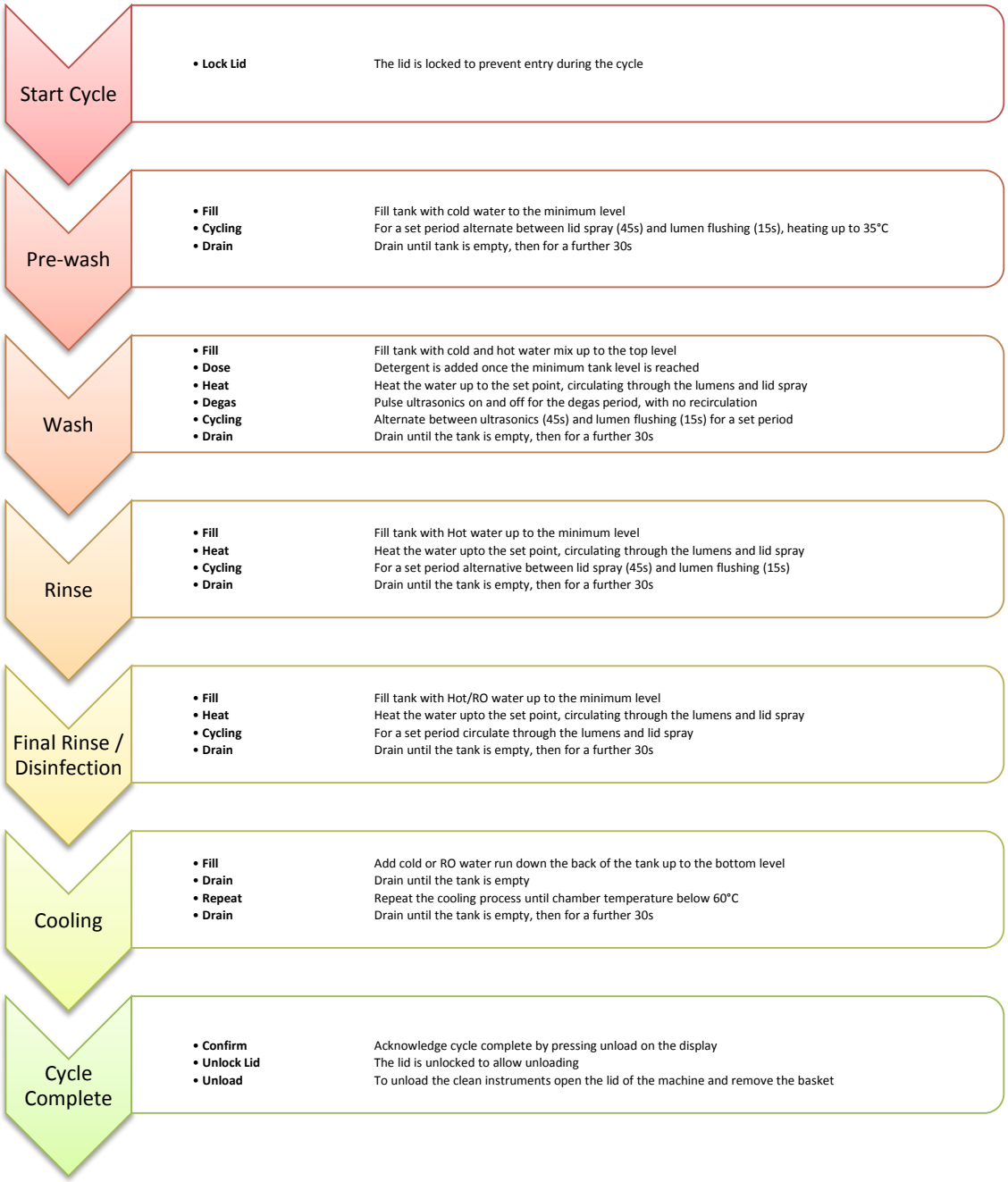
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3-11 Automated Wash Cycle

The automated wash cycle consists of the stages Pre-wash, Wash, Rinse, Final Rinse/Disinfection and Cooling. Every stage of the cycle is configured by the Program Parameters stored in the machines memory (see section 3-5-1). These parameters allow the cycle to be customised to select the fill water type, skip stages, repeat stages, control temperature set points, set stage duration etc.

During a cycle, the remaining time for each stage will be shown on the right hand side of the display, and the temperature/pump pressure on the left.

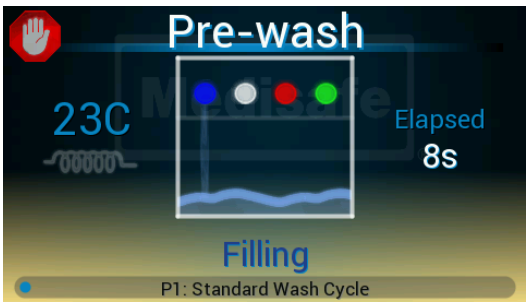
3-11-1 Cycle Process



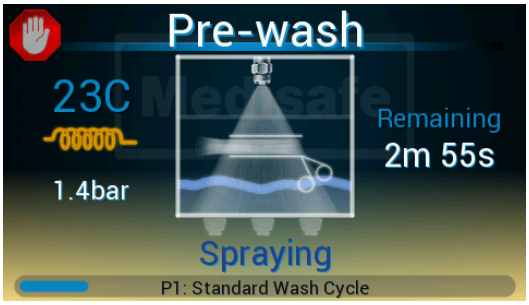
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3-11-2 Pre-wash

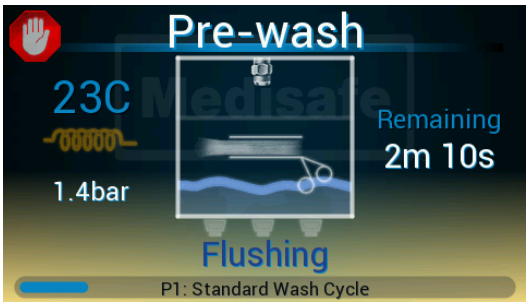
All water in the tank is drained. The machine fills with cold water to the minimum level. During filling for each stage, the spray and irrigation valves are opened, and the pump is operated at least once to ensure all voids are filled.



The machine then washes for a set time, alternating between spraying and flushing.



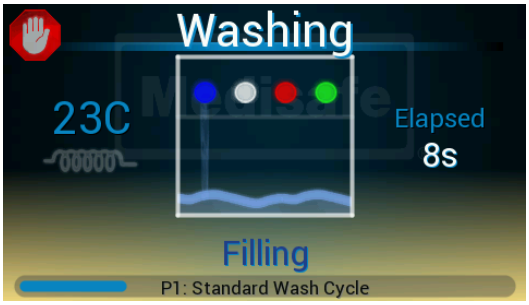
During flushing recirculated water is pumped through the instruments.



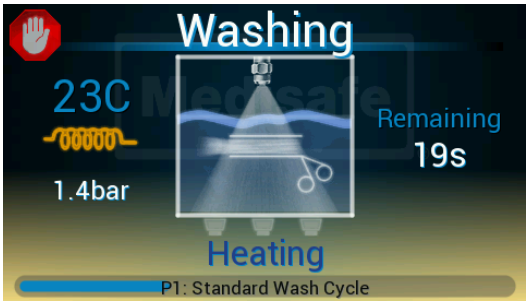
Finally, all water in the tank is drained

3-11-3 Wash

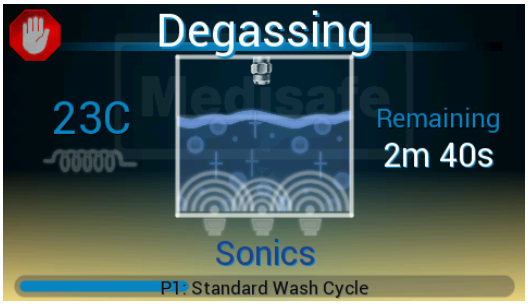
The tank is filled to the top level with a mix of hot and cold water to attain the set temperature. Once the water reaches the mid-level, a quantity of detergent is dispensed into the tank. The pump is pulsed on and off to even out the temperature, fill all cavities, and to mix the chemical thoroughly.



Once the tank is full, if the temperature is below the set point, the water heater will turn on.



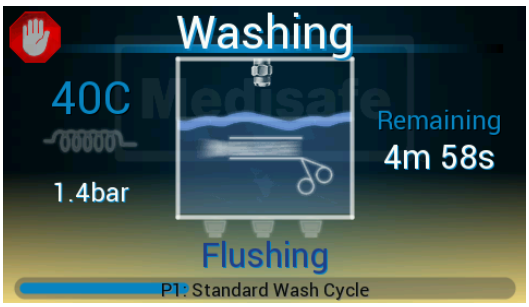
When the set temperature is reached, the ultrasonics degas the tank by pulsing on and off for a pre-defined time to remove any dissolved air.



At the end of the degas period the machine will then alternate between irrigation and *Ultrasonics* for a pre-defined time.

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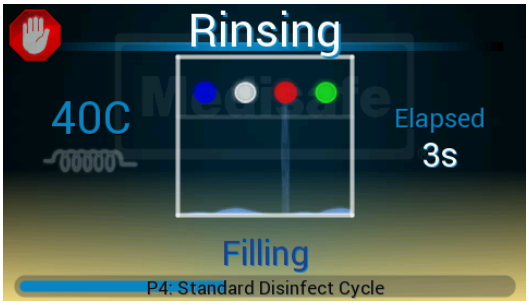




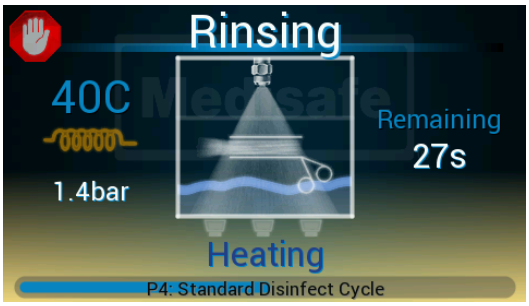
Finally, all water in the tank is drained.

3-11-4 Rinse

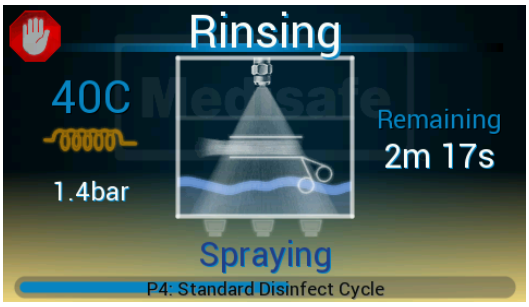
The machine fills with hot water to the minimum level.



Once the tank is filled, the water heater will turn on to heat the water to the set point.



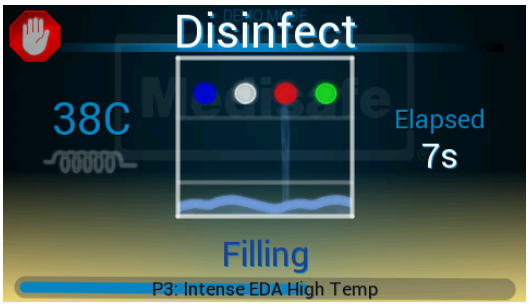
When the set point is reached, the machine will alternate between *Flushing* and *Spraying* for a pre-defined time.



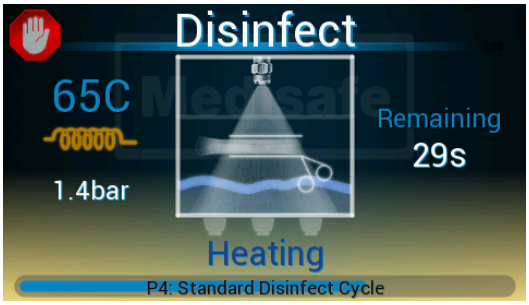
Finally, all water in the tank is drained.

3-11-5 Final Rinse/Disinfection

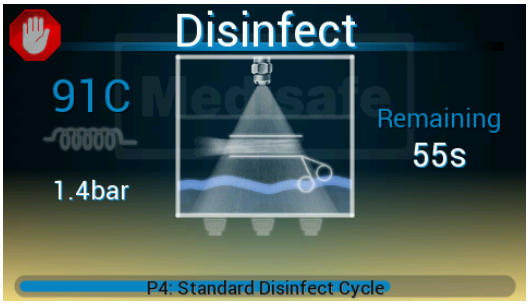
The machine fills with hot water (or RO if fitted) to the minimum level.



Once the tank is filled, the water heater will turn on to heat the water to the set point.



When the set point is reached, the machine will *Flush* and *Spray* for a pre-defined time.

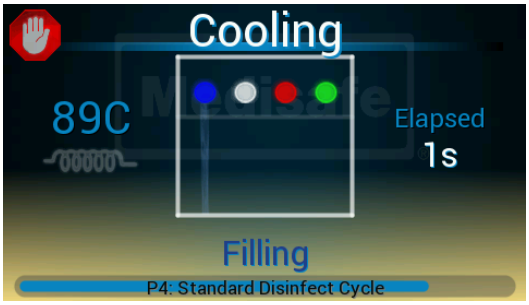


Finally, all water in the tank is drained.

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3-11-6 Cooling

On completion of the last cleaning stage, the inside of the machine will be hot. To reduce the water vapour and temperature, cold water is cascaded down the back wall of the tank and into the drain until the chamber temperature is below 60°C (140°F) or ten minutes have elapsed.

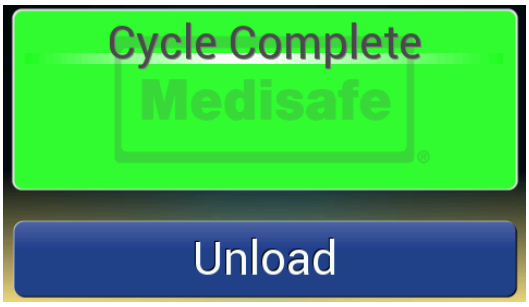


Note: *this water does not come into contact with the load.*

Finally, all water in the tank is drained.

3-11-7 Cycle Complete

When the cycle is complete, a sound is played.



Confirm cycle record is present and indicates the cycle has completed with no errors.

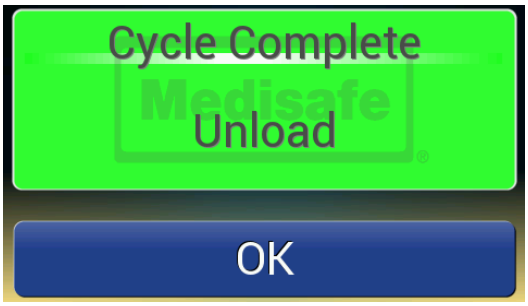
3-11-8 Unload

Open the lid of the machine.

Remove the basket by pulling the handles up to disingage the latch, then lift from the machine.



Press ***OK*** on the screen to confirm unloading complete



Caution

If any instruments became detached from the Medisafe connection tubing during processing, they should be reprocessed ensuring that the Luer connection is firmly seated.

The machine is now ready for the next cycle



3-11-9 Robotic Instrument Preparation

To prepare the processed robotic instruments for disinfection or sterilisation:

1. Dry the instrument completely. Ensure that all water is emptied from the instrument shaft and main flush port by vertically positioning the instrument with tip up while drying. Dry the outside of the instrument with a lint-free cloth. Air may be blown through all flush ports to facilitate drying.
2. Lubricate the tip and wrist mechanism with a neutral pH, steam-permeable instrument lubricant as per the manufacturer's instructions.
3. Proceed to disinfection after program 3 or sterilisation after program 6.

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3-11-10 Re-Wash Cycle

As a water and time saving feature, Program 2 is a re-wash program. This allows the wash water and chemical to be reused. At the end of the wash, the user is allowed to open the lid, add new instruments, and repeat the wash stage using the same water up to ten times.

***Note**: if the water temperature exceeds the set point by more than 10°C the cycle will be aborted (see section 3-12).*

At the end of a wash stage press ***Yes*** on the screen to reuse the water again.

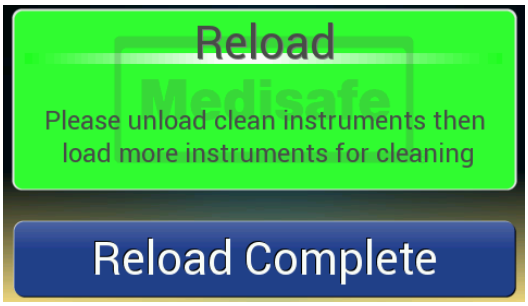


Unload the instruments and load more instruments for cleaning.



Temperature of the water within the unit can reach 60°C (*140°F*), take care when opening lid and reloading the instruments.

Press ***Reload Complete*** on the screen to begin another wash stage



To end the re-wash process, once the wash stage is complete press ***No*** on the screen to drain the water and end the cycle.



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3-12 Cycle Errors

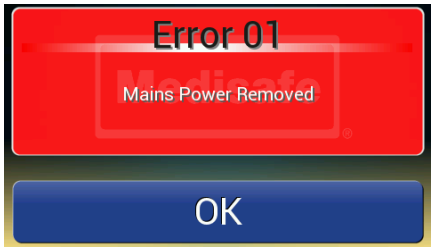
If an unexpected condition occurs during a wash cycle (i.e. low water level, low water temperature, see section 4-1 for a full list) the system will abort the cycle and report an error.

Warning

When an error occurs during a cycle the load may still be contaminated, and will still require processing. Check the printout to see if the cycle completed before the error, if it did not then repeat the cycle to attempt to reprocess the load. If the error persists use another machine or alternative cleaning and disinfection process (according to facility procedure), then contact a service engineer to repair the machine.

If an error occurs:

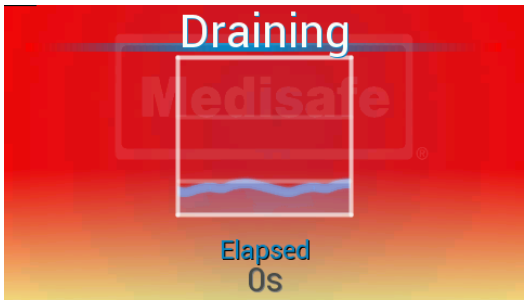
- An audible alarm will sound
- An error code and description will be shown on the display:



- The printer will output the Error code, description, and diagnostic information

Acknowledge the error by pressing ***OK***.

If the tank is not empty, the water will be automatically drained to allow access to the load.



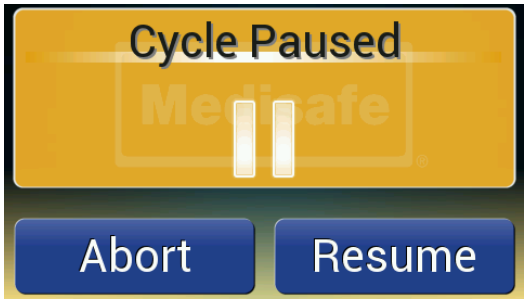
Once the tank is drained, the machine will be reset and return to the main menu.



3-13 Aborting a Cycle

While running a cycle in the SI PCF, it may be necessary to end (abort) the cycle operation before scheduled completion, because of a unit problem, wrong load, etc.

To abort a cycle press in the top left corner. This will pause the cycle, press ***Abort*** to confirm or press ***Resume*** to continue the cycle.



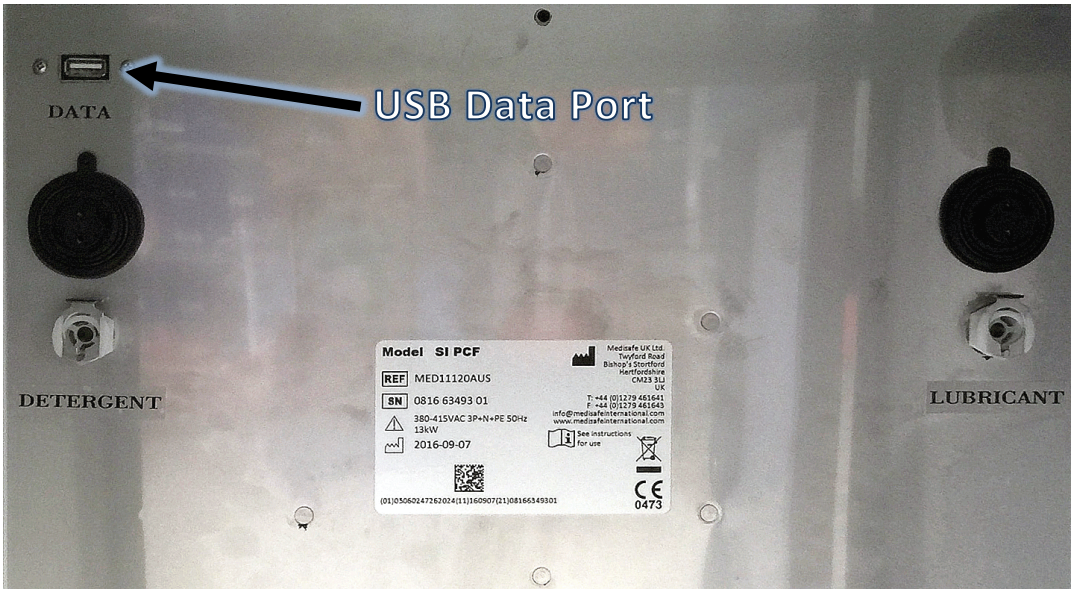
If the cycle is aborted, a cycle error will be generated (see section 3-12)



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3-14 Downloading the Cycle Log

The SI PCF contains on board memory, which holds a record of every cycle run on the machine. This Log can be downloaded to a USB flash drive via the USB Data port located in the chemical cabinet on the right hand side of the machine.



3-14-1 Download Logs

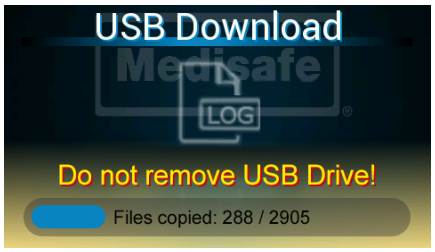
1. Use the scroll ⟨ ⟩ arrows to cycle through to the log icon



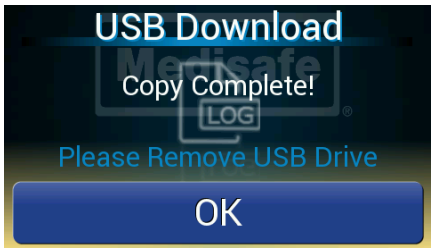
2. Insert the USB flash drive into the USB data port.

   ***Note**: the flash drive must be formatted as FAT32.*

3. Press ***All Logs*** to download all cycle logs, or ***New Logs*** to download the cycle logs since the last download.
4. Wait for the logs to be downloaded to the USB drive. **Do not** remove the USB Drive while downloading, as this could corrupt its contents.



5. Once the log files have been downloaded remove the USB drive and press ***OK***



6. A file for each cycle will be stored in the USB drive directory:

   MEDISAFE\<machine serial number>\<cycle range>\<cycle number>.txt

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3-15 Routine Maintenance

Important safety information:

Always ensure mains power is disconnected.

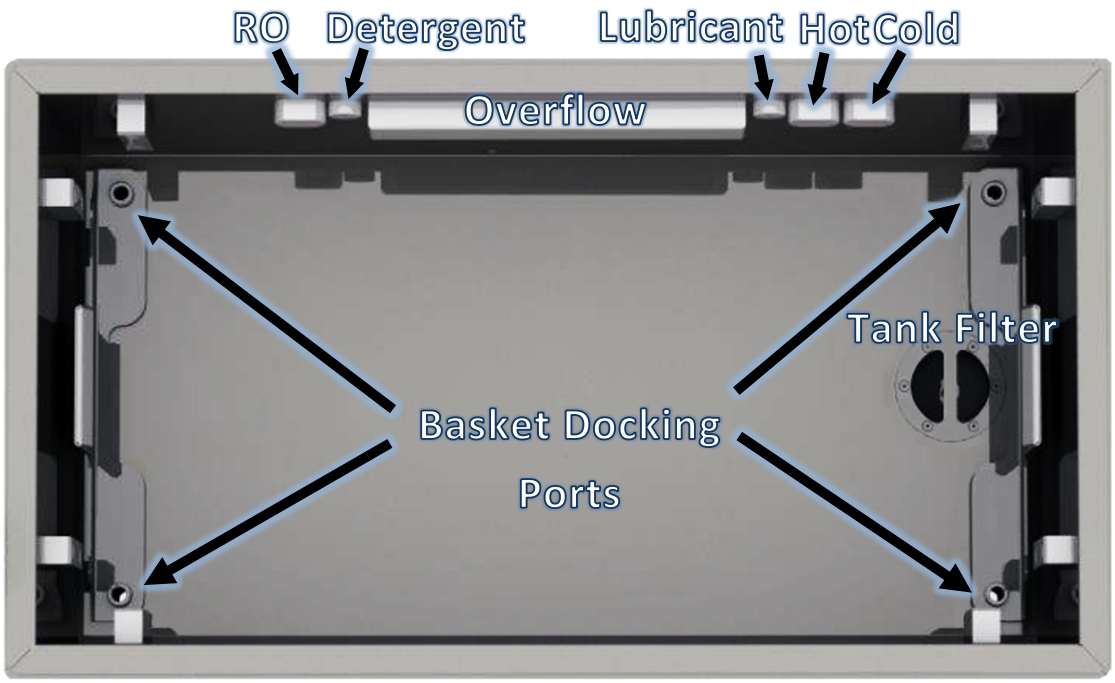
Always adhere to the recommended safety precautions for cleaning products in use.

It is recommended that Personal Protection Equipment (PPE) be worn during loading/unloading, and cleaning or decontaminating the equipment. This must include apron, gloves and eye-protection.

For cleaning internal and external surfaces, a disinfectant cleaner such as Isopropyl Alcohol or a household kitchen cleaner can be used, provided it is marked as a DISINFECTANT cleaner.

In the interests of health and safety, all medical washers should be cleaned on a weekly basis as part of a routine maintenance program. This will ensure the machine is safe for operators and maintenance personnel. To keep your machine operating within its design parameters the following checks must be performed on a regular basis. Please note that completion of these checks is not a substitute for regular servicing and maintenance by a suitably qualified engineer.

All maintenance procedures should be carried out in a well-lit environment, recommended minimum lighting level is 500Lux.



PCF Wash Tank Connections

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3-15-1 Before Every Cycle

1. Visually check that there is enough chemical for the next wash. Replenish as necessary
2. Remove, clean and refit the tank filter, checking the O-ring is still attached and not damaged



3. Check O-rings still attached to the four basket docking ports on the tank



4. Check the wash tank is free from any debris or deposits
5. Check the spray nozzles for blockage
6. Check the condition of basket irrigation port Luer hoses, ensure they are not kinked or ruptured.

Caution If damage is found, please contact a service engineer (see section 1-17).

3-15-2 Daily Checks

1. Automatic control test - select a recent printout and compare parameters with the first printout generated after installation
2. Check that there is sufficient chemical for the days planned use
3. Check and clean all internal surfaces above the water line
4. Check and clean all external surfaces including lid seal
5. Check mains power cord is not trapped, damaged, and the plug is secure in its socket (if fitted)
6. Check lid and lid seal for any damage

Caution If damage is found, please contact a service engineer (see section 1-17).

3-15-3 Weekly checks

1. Carry out the daily checks
2. Check lid opening/closing operation including gas struts
3. Check water hardness of hot and cold water supplies
4. Check water conductivity of final rinse water supply
5. Check condition of mains water supply hoses
6. Visually inspect water and drainage system for leaks
7. Cleaning efficacy test by residual soil detection
8. Run a full decontamination procedure (See section 3-16)

Note: *The use of a "Process Challenge Device for Minimally Invasive Surgery" may be required according to your local authority guidelines, and the cleaning indicator manufacturer's instructions*

Ref: *ISO15883-1:2009*

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3-16 Decontamination Guide

Decontamination is defined as the cleaning and disinfection of the machine, and is vital to ensure the machine is safe for operators and maintenance personnel.

Important safety information:

Always ensure mains power is disconnected.

Always adhere to the recommended safety precautions for cleaning products in use.

Always work in a well-lit area, recommended minimum lighting level is 500Lux.

It is recommended that Personal Protection Equipment (PPE) be worn during loading/unloading, and cleaning or decontaminating the equipment. This must include apron, gloves and eye-protection.

For cleaning internal and external surfaces, a disinfectant cleaner such as Isopropyl Alcohol or a household kitchen cleaner can be used, provided it is marked as a DISINFECTANT cleaner.

3-16-1 Daily Cleaning/ Procedure

Using a clean cloth or paper towel:

1. Wipe down **all** internal surfaces above the waterline with your disinfectant, including the inside of the lid and around the seals
2. Wipe **all** external surfaces, including around the sides of the doors
3. Remove any trace of cleaner from internal and external surfaces with a clean wet cloth
4. Towel dry

***Note**: By the nature of its operation, the washer disinfector's tank is immersed in water during the cleaning/disinfection cycle. Therefore, it should already be clean and disinfected by the end of a cycle.*

3-16-2 Full Decontamination

On a weekly basis:

1. Perform the daily cleaning procedure
2. Fit luers to all basket port to ensure they are open
3. Run Program 6 (see section 3-5 and 3-9)

If the machine has been idle for more than a week:

1. Perform the daily cleaning procedure
2. Fit luers to all basket port to ensure they are open
3. To make a 5% disinfectant solution place 1.2 litres (typically) of non-foaming Sodium Hypochlorite based disinfectant into a shallow heat resistant container. Place the container in the basket (*this prevents it from being drained at the start of the cycle*)
4. Select Program 6 (see section 3-5)
5. Start the cycle (see section 3-9)
6. 30 seconds before the Pre-wash stage finishes, press  to pause the cycle
7. Wait 15 minutes then press ***Resume*** to continue the cycle
8. Allow the cycle to complete

3-16-3 Pre Transport Procedure

In the unlikely event of a machine having to be returned to Medisafe for any reason, the following procedure must be carried out:

1. Drained and completely empty of fluid
2. All external hoses removed
3. Fully decontaminated prior to shipping, (*irrespective of when it was last cleaned*)
4. Accompanied by the relevant signed Decontamination Certificate (see section *3-16-4*)

***Note:** If due to the nature of any fault present, the unit cannot be run prior to return, the tank/chamber must be decontaminated by hand.*

***Note:** Failure to carry out this procedure will incur an additional charge for a full decontamination.*

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3-16-4 Decontamination Certificate

Ensure that all items returned to your supplier are disinfected in accordance with the appropriate hospital procedure for equipment of this type.

When returning this product to your supplier, the form must be completed and attached to the outside of the crate/carton. If it is not visible, the courier and/or your supplier can refuse to handle this product.

DECONTAMINATION CERTIFICATE

I hereby certify that the machine referred to in this document by model number and serial number has been disinfected in accordance with Hospital Procedure

Model No. _____

Serial No. _____

Hospital: _____

Name: _____

Signature: _____ Dated: _____

Medisafe®

Promoting a Clean and Safe Service

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4. Troubleshooting

User Manual



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This section of the manual describes the types of SI PCF problems which may occur, and indicates probable causes/corrections.

If the situation cannot be corrected following the Troubleshooting Solutions, repeating the cycle, or if a problem occurs which is not described, then please contact Medisafe. Service charges may be incurred, consult your warranty for details. A trained service technician will promptly place your SI PCF in proper working order.

NOTE: *Never permit unqualified persons to service the system. If a system problem cannot be corrected using this section, please contact Medisafe.*

Calling for Service: First, try to define the situation and determine whether it can be solved (see section 4-1 and 4-2). If the situation cannot be solved, call Medisafe and provide the following information:

- A description of the fault
- Model number
- Serial number
- A copy of the printout showing the error

4-1 Error Messages

Error Code	Error Message	Possible Causes and Solutions
1	Mains Power Removed	Check mains supply to unit was not interrupted during a cycle
2	Chamber Overflow	Switch the machine off then on again, if the problem persists turn the machine and the water supply off and call a service engineer
3	Drain Timeout	Check tank filter clear, ensure drain is not blocked Check that genuine Medisafe chemical is being used (*EDA+ or 3E-Zyme*®)
4	Chamber Temperature Sensor Mismatch	Repeat cycle
5	Chamber Fill Timeout	Check water supply(s) to the machine are switched on Check water inlet hose(s) for kinks
6	Chamber Opened	Check lid is closed
7	Temperature Set Point Not Maintained	Repeat cycle, if the problem persists call a service engineer
10	Chamber Water Level Low	Check that a non-foaming detergent is being used Check for leaks
11	Chamber Water Heat Rate Low	Repeat cycle Temperature sensor or water heater element failure, please call a service engineer.
13	Chemical Dosing	Check chemical bottle level(s) and replace if required. Check chemical pickup tube is inserted into the chemical container

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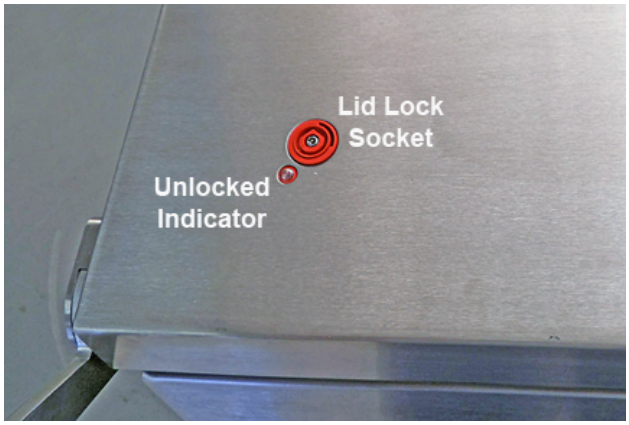
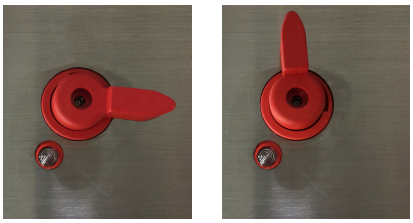
Error Code	Error Message	Possible Causes and Solutions
14	Chamber Start Temperature too High	Check the Cold water temperature is not above 35°C The chamber has not cooled sufficiently since the last wash. Leave more time between washes for the machine to cool down.
15	Chamber Temperature Out of Range	Switch the machine off then on again
16 17 18 19 51	Level Sensor Invalid	Repeat cycle
20	User Aborted Cycle	Repeat cycle
21	Low Main Pump Pressure	Check tank filter is clear/clean Check Basket is docked properly Check Basket ports are not blocked Check for leaks
22	High Main Pump Pressure	Attach a Luer pipe to at least one manifold to aid water circulation Check Basket is docked properly Check Basket ports are not blocked Check for leaks
23	Temperature Set Point Exceeded	Repeat cycle
39 40 41 43 45 46 47	Sonic Generator Fault	Switch the machine off then on again
42	Maximum Cycle Time Expired	Check water supply flow rate is sufficient Check tank filter clean/clear, ensure drain is not blocked Repeat cycle
44	Sonic Cleaning Water Level Low	Check for visual leaks
49 57 58	Microcontroller Error	Switch the machine off then on again
50	Internal Ambient Temperature too High	Switch the machine off then check for blockages in the ventilation holes Switch the machine on again after an hour
52	Real Time Clock Read Error	Switch the machine off then on again
53	24V Power Supply Voltage Out of Range	Switch the machine off then on again
54	Overfill occurred during cool down process	Check tank filter clear, ensure drain is not blocked Repeat cycle
56	Low Wash Irrigation Pressure	Check tank filter is clear/clean Check Basket is docked properly Check Basket ports are not blocked Check for leaks
59	Unable to Store Cycle Log	Repeat cycle

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4-2 Other Faults

Problem	Possible Causes and solutions
Poor Load Cleaning	• Insufficient detergent injected. Contact Medisafe • Check that an approved chemical is being used • Instrument basket is overloaded. Adjust load • Instruments are excessively soiled • Instruments have been left soiled for more than two hours • Verify facility power meets Unit specifications
Water Leaking from Lid Area	• Check instruments are lying flat in the basket and not pointing up at the lid • Check lid seal is undamaged
Excessive Fill Time	• Check incoming water pressure is above 2 bar (29 psi) • Check water inlet valve is not blocked (beware of hot water temperature!)
Low Irrigation Port Flow Rate	• Check all connectors are fully inserted • Clean tank filter • Check basket manifold ports are not blocked • Check tank docked ports are not blocked • Verify facility power meets Unit specifications
No Power to Unit or Display is off	• Installation Main disconnect switch is in the OFF position Turn main disconnect switch to the ON position • Facility power is incorrect for unit. Correct facility power to meet unit requirements • Internal Circuit breaker has tripped, contact Medisafe

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Problem	Possible Causes and solutions
Lid will not open	• Installation Main disconnect switch is in the OFF position Turn main disconnect switch to the ON position • To override the lid lock: The lid lock override socket is on the top left of the machines lid:  Insert the key into the lid lock socket and turn anticlockwise to override the lock to allow the lid to be opened, as shown below.  Normal Override **Warning** **The override key must only be used to gain access when the lid is locked and cannot be opened normally, and the machine is not running a cycle**. **Warning** In the case of an incomplete cycle the load will require to be decontaminated, see section 3-12

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5. Technical
User Manual



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5-1 General Specifications

General	
Description	Medical Device Class IIb
Intended Purpose	Internal and external cleaning of surgical instruments, hollow-ware, and associated products
Operating Process	Ultrasonic activity, Spray Ports and Irrigation Ports
Classifications	Class 1 electric shock protection, externally powered No applied parts IP rating – IPX0 (refer to BS EN 60601-1, para 6.3) Not suitable for use in an oxygen rich environment Mode of operation – intended for continuous use
Lid	Locking lid prevents access to wash chamber during cycle
Deluge Wash	Top Spray in lid
Irrigation	With ten ports open each basket manifold outputs produces up to 3 bar (43.5psi) pressure with a flow rate of 50ml/s
Basket Connection	Drop and Lock for easy irrigation connection

User Interface	
Display	TFT Touch Screen

Standard Basket	
Features	Four 5-way manifolds, one in each corner
Din Basket Compatibility	Full DIN Tray Standard to hold instruments (M20044) – Length 484mm (19in) Width 253mm (10in) Height 76mm (3in)
Max Load Size	Length 668mm (26.3in) Width 338mm (13.3in) Height 76mm (3in)
Max Load Weight	10kg / 22lb

Si Robotic Basket	
Features	Four 3-way manifolds, one in each corner
Robot Instrument Capacity	10

Xi Robotic Basket	
Features	Two 3-way and Two 2-way manifolds, one in each corner
Robot Instrument Capacity	6

Ultrasonics	
Frequency	34 to 40 kHz dependant on load
Transducers	20 piezo ceramic
Power	800W
Features	Pulsed Degas
	Auto-tune
	Frequency Sweep
	Reflectasonic Technology

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Electrical Supply	
General	Mains supply voltage fluctuations up to ±10% of the nominal voltage; Transient over voltages typically present on the mains supply (Overvoltage Category II). Pollution Degree 2
MED11100 (Standard)	Voltage: 200 - 230VAC (*Single Phase + Protective Earth)* Frequency: 50Hz Power: 5kW Site Current Capacity: ≥ 18A per phase Site MCB and Disconnect Switch Current Rating: 25A Site Wiring Cross-sectional area: 4mm^2
MED11110 (Standard)	Voltage: 200 - 230VAC (*Single Phase + Protective Earth)* Frequency: 60Hz Power: 5kW Site Current Capacity: ≥ 18A per phase Site MCB and Disconnect Switch Current Rating: 25A Site Wiring Cross-sectional area: 4mm^2
MED11122 (Standard) MED11130 (Robotic)	Voltage: 200 - 230VAC 3P+PE (*Three Phase + Protective Earth*) Frequency: 50Hz Power: 10kW Site Current Capacity: ≥ 24A per phase Site MCB and Disconnect Switch Current Rating: 32A Site Wiring Cross-sectional area: 6mm^2
MED11121 (Standard) MED11121UL (Standard) MED11131 (Robotic)	Voltage: 200 - 230VAC 3P+PE (*Three Phase + Protective Earth*) Frequency: 60Hz Power: 10kW Site Current Capacity: ≥ 24A per phase Site MCB and Disconnect Switch Current Rating: 32A Site Wiring Cross-sectional area: 6mm^2
MED11120 (Standard) MED11132 (Robotic)	Voltage: 380 - 415VAC 3P+N+PE (*Three Phase + Neutral + Protective Earth*) Frequency: 50Hz Power: 13kW Site Current Capacity: ≥ 18A per phase Site MCB and Disconnect Switch Current Rating: 25A Site Wiring Cross-sectional area: 4mm^2
MED11123 (Standard) MED11133 (Robotic)	Voltage: 380 - 415VAC 3P+N+PE (*Three Phase + Neutral + Protective Earth*) Frequency: 60Hz Power: 13kW Site Current Capacity: ≥ 18A per phase Site MCB and Disconnect Switch Current Rating: 25A Site Wiring Cross-sectional area: 4mm^2

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Water Supply	
Back Siphonage Protection	Integral AB type air gap and weir observing BS EN 1717
Consumption (approximate)	Per cycle stage: • Pre-wash 16lts (4.2US gal) • Wash 50lts (13.2US gal) • Rinse 16lts (4.2US gal) • Disinfection 16lts (4.2US gal) • Cooling 20lts (5.3US gal) Total Per supply: • Cold 41lts (11US gal) • Hot 41lts (11US gal) • RO 36lts (10US gal) Total Cycle 118lts (32US gal) Consumption will vary depending on the cycle selected and the site facilities

Drainage System	
Type	Pumped
Drain Connection	19mm (¾*in*) hose tail outlet
Overflow	48.3mm (1.9*in*) OD outlet
Effluent Temp	95°C (*203°F*)

Dimensions	
Wash Chamber Liquid Volume	43 Litres (*11.5US gal*)
Wash Chamber Capacity	69 Litres (*18.2US gal*)
Wash Chamber Dimensions	Width 750mm (29.5*in*)
	Depth 365mm (*14.4in*)
	Height 237mm (*9.3in*)
Outer Cabinet Dimensions	Width 876mm (*34.5in*)
	Depth 669mm (26.*3in*)
	Height Lid Closed 1123mm (*44.2in*)
	Height Lid Open 1490mm (*58.7in*)
Shipping Container Dimensions	Width 1060mm (*41.7in*)
	Depth 880mm (*34.6in*)
	Height 1280mm (*50.4in*)

Weight	
Operational weight (*Dry*)	150kg (*331lb*)
Shipping weight	217kg (*479lb*)

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Chemicals	
Storage	Integral Chemical cupboard for 2 bottles
Size and Nature of Bottles	EDA+ or 3EZyme Fluid: 10L (2.6US gal) plastic bottle
Quantity Used per Cycle	EDA+:360ml (12fl.oz US) or 3EZyme:360ml (12fl.oz US)

Environmental conditions	
Use	Indoor, Altitude up to 2000m (*6562ft*)
Ambient Operating Temperature	5°C - 40°C (*41°F - 104°F*)
Humidity	Maximum relative humidity 80% for temperatures up to 31°C (*87.8°F*) decreasing linearly to 50% relative humidity at 40°C (*104°F*)
Noise	Max SPL 68dBA @ 1m from the front of the machine

Real Time Clock Battery	
Type	CR1220 (5012LC)
Typical Capacity	35 - 40 mAh
Standard Discharge Current	0.1 mA
Dimensions	12.5mm x 2.0mm

Range and Accuracy	
Water Temperature	Measurement range 2Cº to 98Cº (36ºF to 208ºF) Displayed and printed to the nearest ºC or ºF accuracy ±0.5ºC or ±0.9ºF Max water temperature: 95ºC (limited by software and thermal trip)
Time	Displayed in minutes and seconds, accuracy ±0.5sec

Internal Data Storage	
Type	Micro SD FLASH Card
Typical Capacity	4GB (1 Million cycles)

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5-2 Applied Standards

Standard	Description
BS EN ISO 15883-1:2009 +A1:2014	Washer-disinfectors. General requirements, terms and definitions and test
MDD 93/42/EEC	Medical Device Directive
BS EN ISO 15883-2:2009	Washer-disinfectors - Requirements and tests for washer-disinfectors employing thermal disinfection for surgical instruments, anaesthetic equipment, bowls, dishes, receivers, utensils, glassware, etc.
BS EN 61010-1:2010	Safety requirements for electrical equipment for measurement, control, and laboratory use
BS EN 61010-2-040:2015	Particular requirements for sterilizers and washer-disinfectors used to treat medical materials
BS EN 60601-1-6:2010+A1:2015	Medical electrical equipment. General requirements for basic safety and essential performance
BS EN 60601-1:2006 +A12:2014	Medical electrical equipment. General requirements for basic safety and essential performance
BS EN 60601-1-2:2007	Medical electrical equipment. General requirements for basic safety and essential performance. Collateral Standard. Electromagnetic disturbances. Requirements and tests
BS EN 62366-1:2015	Medical devices. Application of usability engineering to medical devices
BS EN ISO 13485:2012	Medical devices - Quality management systems
BS EN ISO 14971:2012	Medical devices - Application of risk management to medical devices
PD IEC/TR 60878:2015	Graphical symbols for electrical equipment in medical practice
BS EN 1717:2000	Protection against pollution of potable water in water installations and general requirements of devices to prevent pollution by backflow
PD 5304:2014	Guidance on the safe use of machinery
EN 61326: 2013	Electrical equipment for measurement, control and laboratory use. EMC requirements. General requirements
UL 61010-1-2nd Edition	Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use
2003/10/EC	Minimum health and safety requirements regarding the exposure of workers to the risks arising from physical agents (noise)
2009/104/EC	Minimum safety and health requirements for the use of work equipment by workers at work

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6. Appendices

User Manual



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EMC

6-1 EMC & Electromagnetic Compatibility Data

Instructions/Notes regarding the SI PCF and Electromagnetic compatibility (EMC) testing to EN60601-1-2:2007

The SI PCF has been tested regarding its ability to operate in an environment containing other electrical/electronic equipment (including other medical devices).

The purpose of this testing is to ensure that the SI PCF is not likely to adversely affect the normal operation of other such equipment and that other such equipment is not likely to adversely affect the normal operation of the SI PCF.

Despite the testing of the SI PCF that has been undertaken, normal operation can be affected by other electrical or electronic equipment. Portable and mobile RF communications equipment can affect medical electrical equipment.

As the SI PCF is medical equipment, special precautions are required regarding EMC.

It is important that the SI PCF is configured and installed/put into service, in accordance with the instructions/guidance provided herein and is used only in the configuration as supplied.

The SI PCF may be supplied with the following accessories: serial data cable (MED1017) and thermal printer (MED1014.1)

The SI PCF's data cable must be restricted to less than 3m in length and should not be extended by the user. If the SI PCF is used with adapted cables or cables other than those supplied, this may result in increased emissions or decreased immunity of the SI PCF, in relation to EMC performance.

If the SI PCF's is used with a printer other than the one supplied this this may result in increased emissions or decreased immunity of the SI PCF, in relation to EMC performance.

It should be noted that the cables provided with the SI PCF should not be used on other equipment. To do so may result in increased emissions or decreased immunity of the other equipment in relation to EMC performance.

The SI PCF should not be used adjacent to or stacked with other equipment. If adjacent or stacked use with other equipment is necessary, the SI PCF and the other equipment should be observed / monitored to verify normal operation in the configuration in which it will be used.

For the purposes of EN60601-1-2, the SI PCF has an essential performance as follows:

Functions within the operating cycle should continue to operate as normal, and the product should complete its operating cycle correctly.

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6-3 Electromagnetic Emissions

Guidance and Manufacturer's Declaration – Electromagnetic Emissions		
The SI PCF is intended for use in the electromagnetic environment specified below. The customer or the user of the SI PCF should assure that it is used in such an environment.		
Emissions test	**Compliance**	**Electromagnetic environment - guidance**
RF emissions CISPR 11	Group 1	The SI PCF uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR 11	Class A	The SI PCF is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Harmonic emissions IEC 61000-3-2	Not Applicable	
Voltage fluctuations / flicker emissions IEC 61000-3-3	Not Applicable	

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6-4 Electromagnetic Immunity

Guidance and Manufacturer's Declaration – Electromagnetic Immunity			
The SI PCF is intended for use in the electromagnetic environment specified below. The customer or the user of the SI PCF should assure that it is used in such an environment.			
Immunity test	**IEC 60601 test level**	**Compliance level**	**Electromagnetic environment - guidance**
Electrostatic discharge (ESD) IEC 61000-4-2	± 6 kV contact ± 8 kV air	± 6 kV contact ± 8 kV air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient / burst IEC 61000-4-4	± 2 kV for power supply lines ± 1 kV for input / output lines	± 2 kV for power supply lines Input / output line tests are not applicable.	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	± 1 kV line(s) to line(s) ± 2 kV line(s) to earth	± 1 kV differential mode ± 2 kV common mode	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	<5% U_T (>95 % dip in U_T) For 0.5 cycle 40% U_T (60 % dip in U_T) for 5 cycles 70 % U_T (30 % dip in U_T) for 25 cycles <5% U_T (>95 % dip in U_T) For 5 s	Not Applicable (Rated input current >16A per phase) Not Applicable (Rated input current >16A per phase) Not Applicable (Rated input current >16A per phase) <5% U_T (>95 % dip in U_T) For 5 s	Mains power quality should be that of a typical commercial or hospital environment. If the user of the SI PCF requires continued operation during power mains interruptions; it is recommended that the SI PCF be powered from an uninterruptable power supply or a battery.
Power frequency (50/60Hz) Magnetic field IEC 61000-4-8	3 A/m	3 A/m (50Hz only)	If incorrect operation occurs, it may be necessary to position the SI PCF further from sources of power frequency magnetic fields or to install magnetic shielding. The power frequency magnetic field should be measured in the intended installation location to assure that it is sufficiently low.
Note: UT is the A.C. mains voltage prior to application of the test level.			

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Guidance and Manufacturer's Declaration – Electromagnetic Immunity			
The SI PCF is intended for use in the electromagnetic environment specified below. The customer or the user of the SI PCF should assure that it is used in such an environment.			
Immunity test	**IEC 60601 test level**	**Compliance level**	**Electromagnetic environment - guidance**
			Portable and mobile RF communications equipment should be used no closer to any part of the SI PCF, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance (d)
Conducted RF IEC 61000-4-6	3 Vrms 150 kHz to 80 MHz	3 Vrms 150 kHz to 80 MHz	d = 1.2√P
Radiated RF IEC 61000-4-3	3 V/m 80 MHz to 2.5 GHz	3 V/m 80 MHz to 2.5 GHz	d = 1.2√P 80 MHz to 800 MHz d = 2.3√P 800 MHz to 2.5 GHz Where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in metres (m). Fields strengths from fixed RF transmitters, as determined by an electromagnetic site survey (*see **Note a***) should be less than the compliance level in each frequency range (see ***Note b***) Interference may occur in the vicinity of equipment marked with the following symbol: 
Note 1: At 80 MHz and 800 MHz, the higher frequency range applies. ***Note 2:*** These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			
a. Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the SI PCF is used exceeds the applicable RF compliance level above, the SI PCF should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the SI PCF. **b.** Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.			

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6-5 Separation Distances Between SI PCF and Mobile RF Equipment

Recommended separation distances between portable and mobile RF communications equipment and the SI PCF

The SI PCF is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the SI PCF can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the SI PCF as recommended below, according to the maximum output power of the communications equipment.

Rated maximum output power of transmitter W	Separation distance according to frequency of transmitter m		
	150 KHz to 80 MHz $d = 1.2\sqrt{P}$	80 MHz to 800 MHz $d = 1.2\sqrt{P}$	800 MHz to 2.5 GHz $d = 2.3\sqrt{P}$
0.01	0.12	0.12	0.23
0.1	0.38	0.38	0.73
1	1.2	1.2	2.3
10	3.8	3.8	7.3
100	12	12	23

For transmitters rated at a maximum output power not listed above, the recommended separation distance d in metres (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.

Note 1: At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.

Note 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures objects and people.

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6-6 Warranty

To ensure continued efficiency and long operational life, your new SI PCF is covered by a one-year warranty/5,000 cycle warranty which is arranged and administered by Medisafe (UK) Ltd. Details of the warranty are explained in this section.

Subject to the following terms and conditions, Medisafe (UK) Ltd. warrants that any item of THE BASIC WARRANTY ITEMS of the unit (hereinafter called “new SI PCF”) manufactured by Medisafe (UK) Ltd. shall be free from any defects in material and workmanship under normal use and maintenance.

Any item of THE BASIC WARRANTY ITEMS covered by the limited warranty will be repaired or replaced free of charge by Medisafe (UK) Ltd. or any distributor authorised by Medisafe (UK) Ltd. to act as their agent (hereinafter called “authorised Medisafe dealer or distributor”).

The Basic Warranty Items

THE BASIC WARRANTY ITEMS are any items (except THE SPECIFIC EXCLUDED ITEMS) of this new SI PCF which are warranted by an authorised Medisafe dealer or distributor within 12 months or 5,000 cycles (whichever occurs first) of the WARRANTY START DATE when this new SI PCF is first installed by, or for you (the original purchaser) as set forth on the Memorandum of Registration.

THE BASIC WARRANTY ITEMS listed below are warranted within 12 months or 5,000 cycles (whichever occurs first) from the WARRANTY START DATE against defects in manufacture.

- Solenoid valves
- Electric motors and pumps
- Switches
- Electronic control circuits
- Water heating elements
- Ultrasonic transducers

The Specific Excluded Items

THE SPECIFIC EXCLUDED ITEMS are any items of this new SI PCF which are not included in THE BASIC WARRANTY ITEMS and are therefore not subject to the terms and conditions of the PRODUCT WARRANTY.

Expendable items, which will deteriorate and require replacement with normal use of the Washer (for example: lid seal, hose and luers, CPE ports, O-rings, various types of gaskets and hoses, LEDs (if fitted), and other expendable parts not specifically listed).

Fair wear and tear of any part, normal maintenance service items, and the parts and any material used in connection with the new SI PCF.

Normal deterioration or damage to plated parts, paint coat, and rubber parts, caused by daily wear and tear and exposure to atmosphere and environment.

Minor irregularities not affecting quality, performance or function of the new SI PCF parts thereof (e.g. slight noise or vibration) or which appear only under particular abnormal conditions.

Damage caused by insufficient or improper maintenance (e.g. negligence of the daily, weekly, and monthly periodic inspections and any servicing requirements as described and specified within this manual).

Damage caused by using a part or lubricants not approved by an authorised Medisafe dealer or distributor or being repaired or replaced by a person who is not an authorised Medisafe dealer or distributor.

Damage caused by accident, abuse, or negligence of the proper handling as set out in this Owner’s Manual, misuse of the new SI PCF, use of the new SI PCF under unusual conditions, and modifications of or to the new SI PCF or any component/parts thereof not recommended or approved by an authorised Medisafe dealer or distributor.

Damage caused by external influences such as chemical spillage, accident, fire, and disasters due to human fault, negligence or Act of God.

Damage to the external casing, wash chamber, basket, or any part of the new SI PCF which may come into contact with the basket during the process of loading/ unloading the new SI PCF.

All charges incidental to breakdowns, such as loss of use of the new SI PCF, loss of time, inconvenience, telephone, transportation, loss or damage to personal property or loss of revenues, or other matter not specifically included.

Damage caused by incorrect installation of the new SI PCF, which could cause operational failure or impairment either immediately after such installation or at any time within the period of this PRODUCT WARRANTY.

Medisafe (UK) Ltd. reserves the right of final decision on all manufacturers’ warranty claims.

Nothing herein affects the statutory rights of the consumer.

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6-7 Service Record

Medisafe recommend that your SI PCF be serviced every 3 months in accordance with the schedule contained within the SI PCF Service Manual, which also contains the technical description of the machine. This record must be completed by the engineer conducting the service and should be made available in the event of a warranty claim or breakdown repair.

Note: *No user serviceable parts inside. Service and repair should be referred to trained and qualified persons only - the product supplier (or Medisafe UK Ltd) to avoid personal injury, improper equipment performance, invalidation of the equipment warranty or other costly damage.*

3 Month		
Serviced by/ Date		
Cycles since last service		
Next service due	Cycles	Date

6 Month		
Serviced by/ Date		
Cycles since last service		
Next service due	Cycles	Date

9 Month		
Serviced by/ Date		
Cycles since last service		
Next service due	Cycles	Date

12 Month		
Serviced by/ Date		
Cycles since last service		
Next service due	Cycles	Date

15 Month		
Serviced by/ Date		
Cycles since last service		
Next service due	Cycles	Date

18 Month		
Serviced by/ Date		
Cycles since last service		
Next service due	Cycles	Date

21 Month		
Serviced by/ Date		
Cycles since last service		
Next service due	Cycles	Date

24 Month		
Serviced by/ Date		
Cycles since last service		
Next service due	Cycles	Date

27 Month		
Serviced by/ Date		
Cycles since last service		
Next service due	Cycles	Date

30 Month		
Serviced by/ Date		
Cycles since last service		
Next service due	Cycles	Date

33 Month		
Serviced by/ Date		
Cycles since last service		
Next service due	Cycles	Date

36 Month		
Serviced by/ Date		
Cycles since last service		
Next service due	Cycles	Date

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6-8 Operator PIN Number Codes

Operator #	Pin	Operator #	Pin	Operator #	Pin	Operator #	Pin
1	2040	21	4060	41	6080	61	8020
2	2181	22	4121	42	6141	62	8161
3	2262	23	4282	43	6222	63	8242
4	2363	24	4383	44	6323	64	8343
5	2484	25	4424	45	6444	65	8464
6	2545	26	4565	46	6585	66	8525
7	2626	27	4646	47	6666	67	8686
8	2727	28	4747	48	6776	68	8787
9	2848	29	4868	49	6888	69	8828
10	2989	30	4929	50	6949	70	8969
11	3070	31	5090	51	7030	71	9050
12	3171	32	5191	52	7131	72	9151
13	3292	33	5232	53	7252	73	9272
14	3353	34	5373	54	7393	74	9333
15	3434	35	5454	55	7474	75	9494
16	3535	36	5555	56	7575	76	9995
17	3656	37	5676	57	7696	77	9636
18	3797	38	5737	58	7757	78	9777
19	3878	39	5898	59	7838	79	9858
20	3979	40	5999	60	7939	80	9959

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6-9 Symbols Glossary

CE 0473	**Conformité Européenne**: A conformity marking for certain products sold within the European Economic Area (EEA) since 1985. The CE marking is also found on products sold outside the EEA that are manufactured in, or designed to be sold in, the EEA. The 4-digit number below the CE mark is the identification number of the notified body involved in the conformity assessment procedure.
	**UL (Underwriters Laboratories)**: An American worldwide safety consulting and certification company
	**WEEE directive:** Shows that a product can be recycled separately from other household waste under the WEEE directive
	Protective Earth symbol
	Standard Hazard Label (*used for Warnings and Hazards*)
	Standard Electrical Danger Label
	Mandatory Read - Read this manual before operating this machine

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